

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

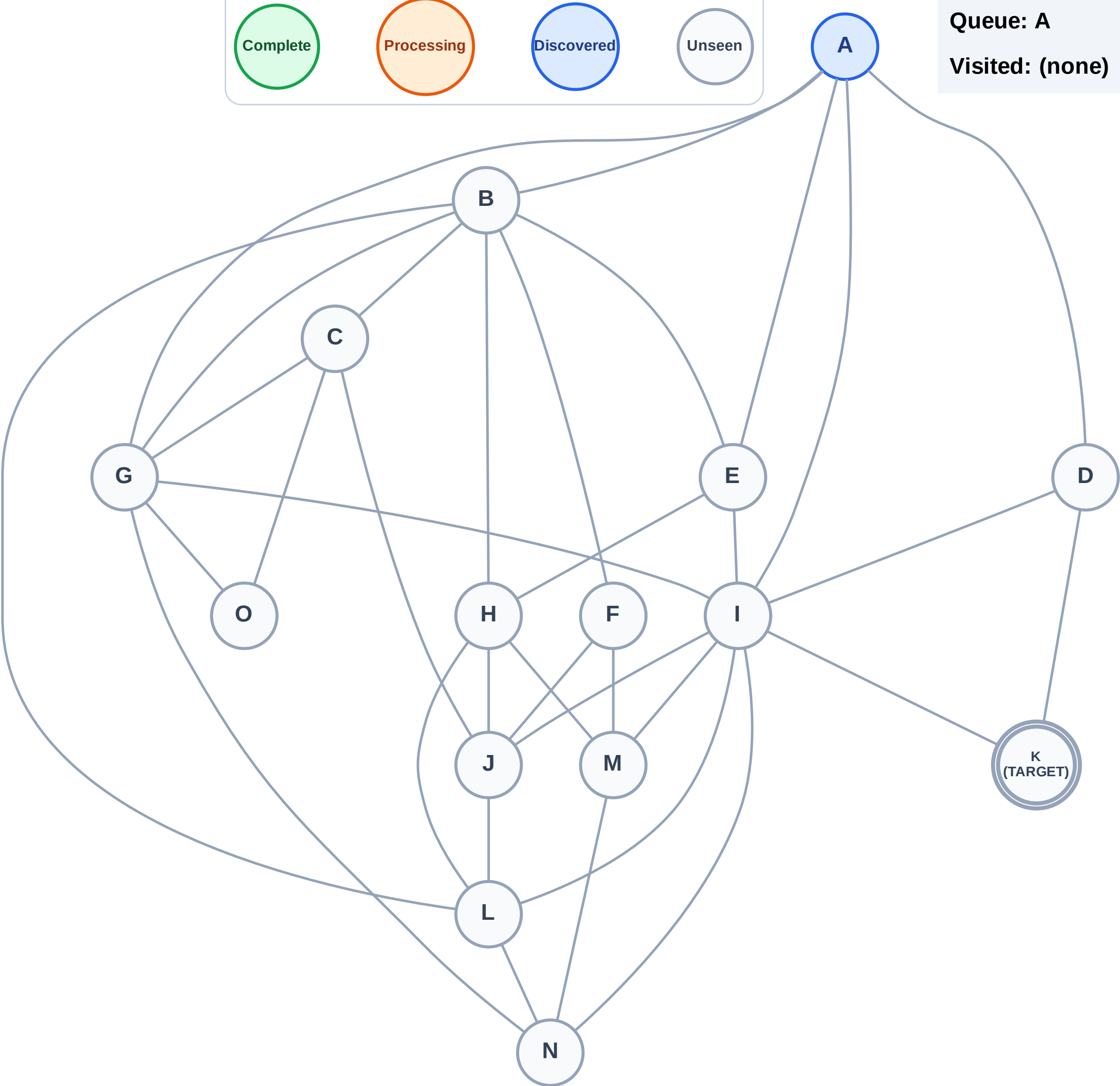
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

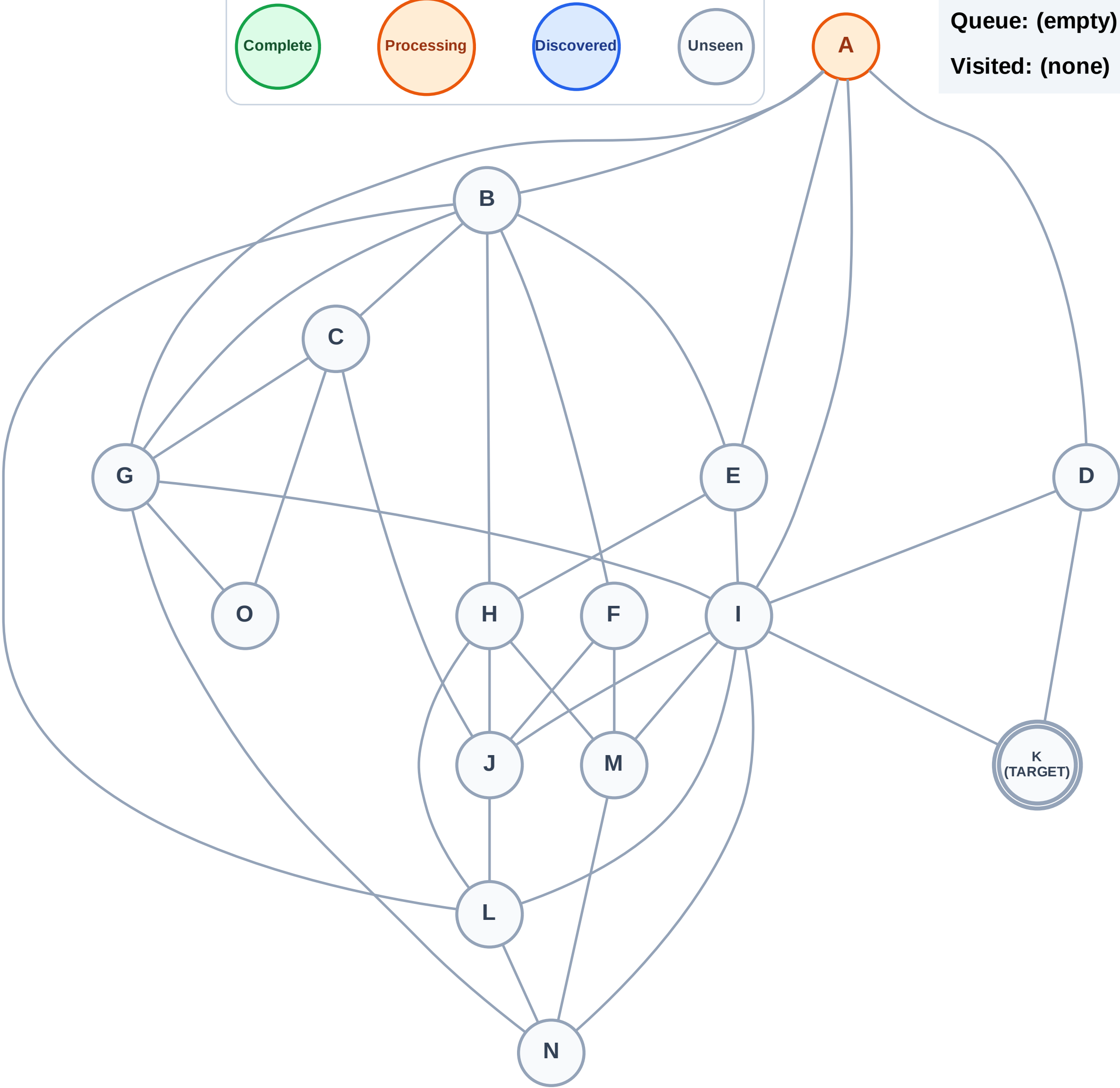
Complete

Processing

Discovered

Unseen

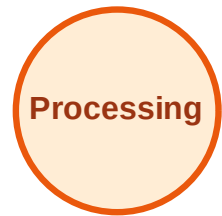
Queue: (empty)
Visited: (none)



BFS Traversal

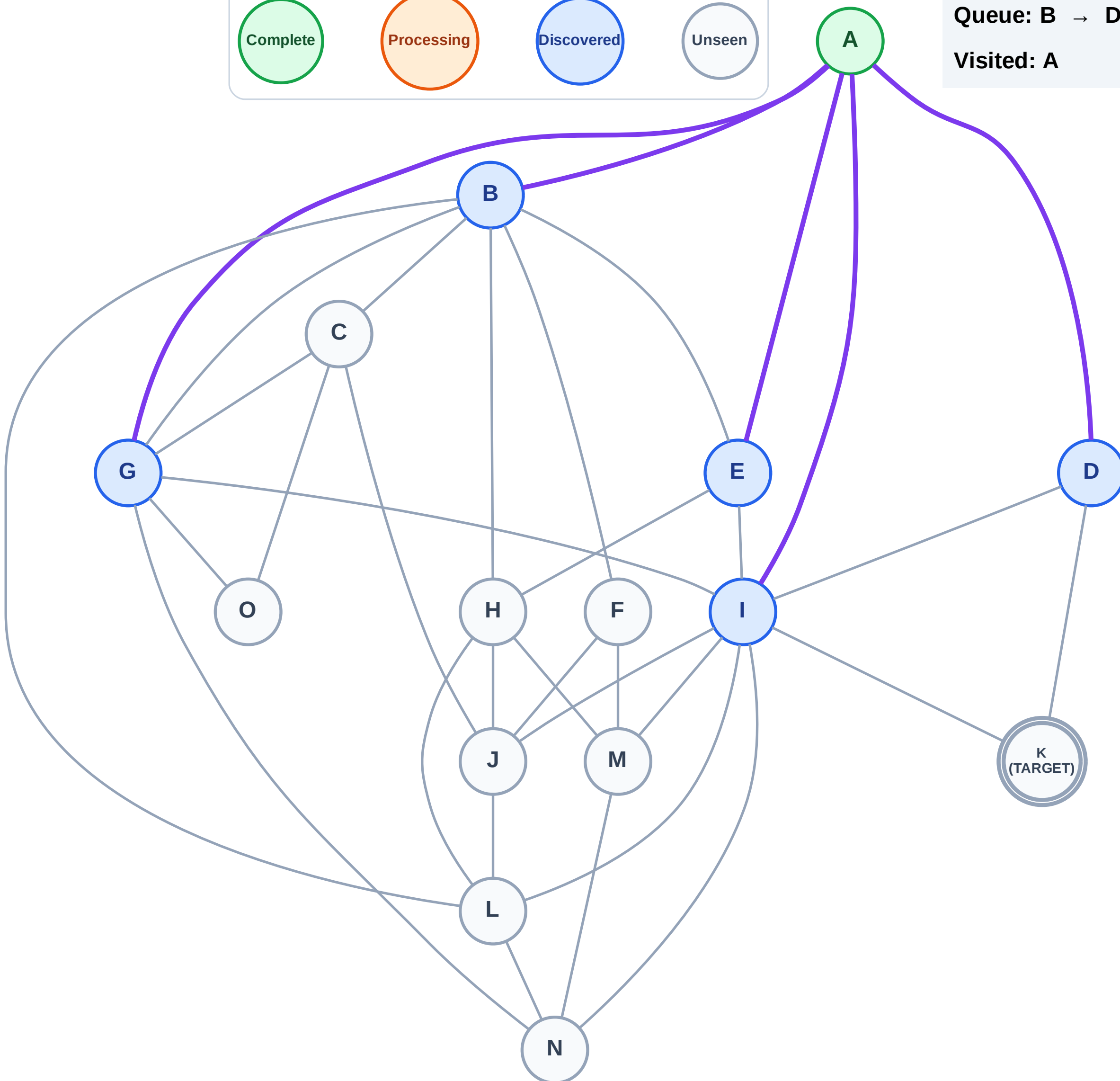
Step 3: Discover B, D, E, G, I from A

Legend



Queue: B → D → E → G → I

Visited: A



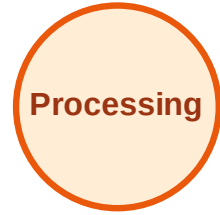
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



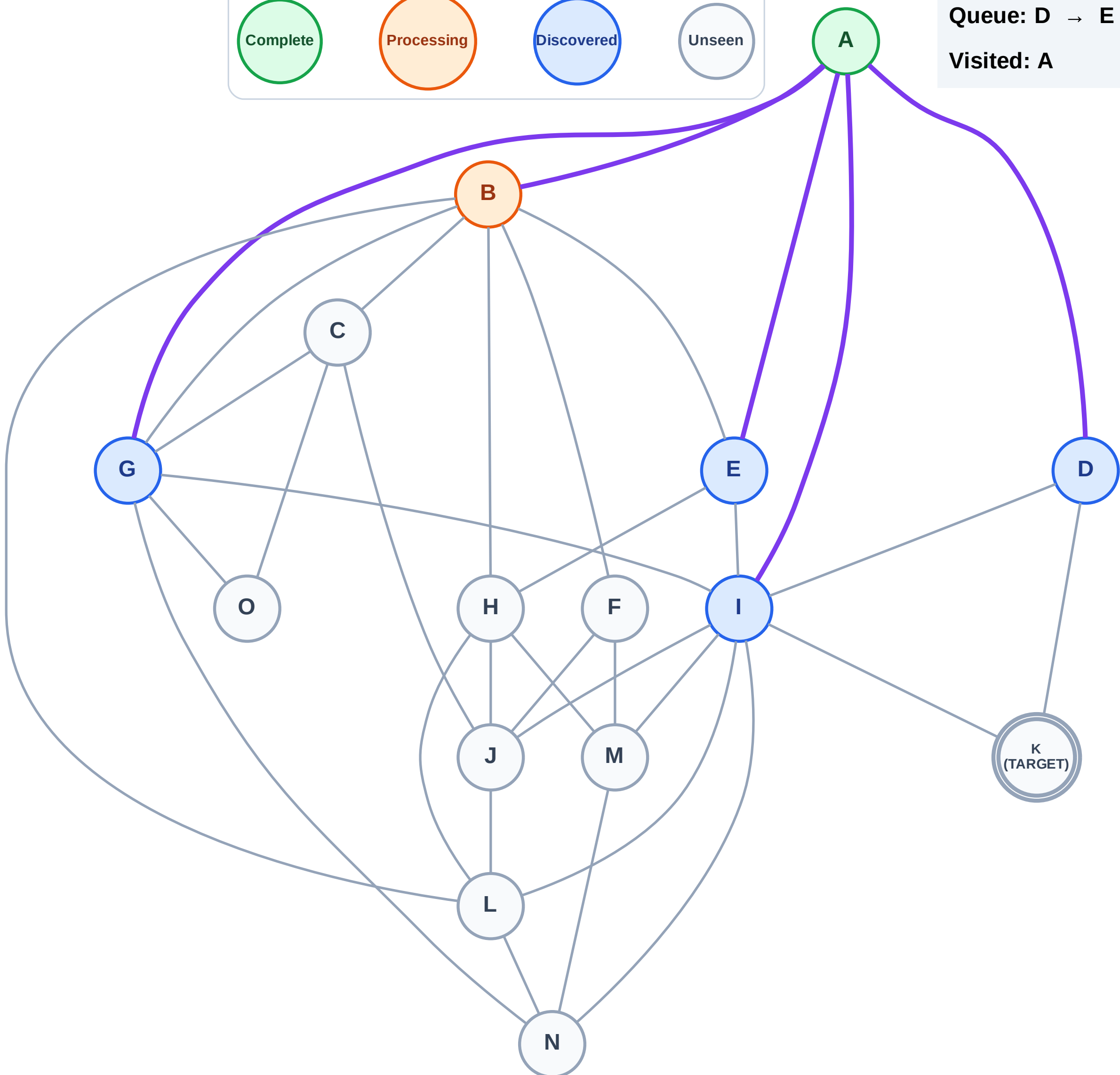
Discovered



Unseen

Queue: D → E → G → I

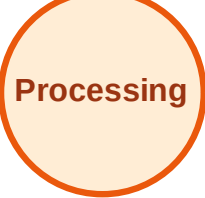
Visited: A



BFS Traversal

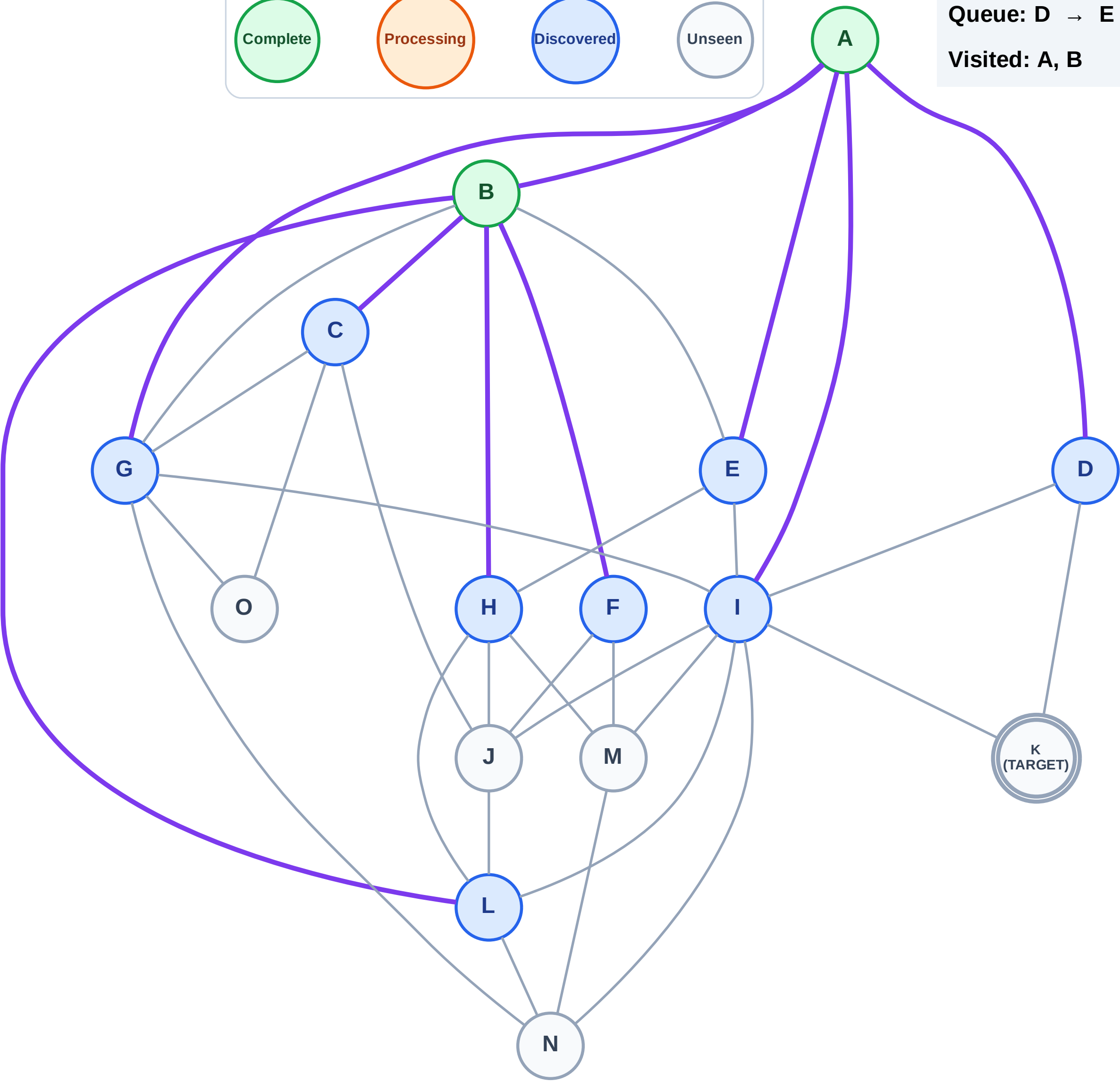
Step 5: Discover C, F, H, L from B

Legend



Queue: D → E → G → I → C → F → H → L

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

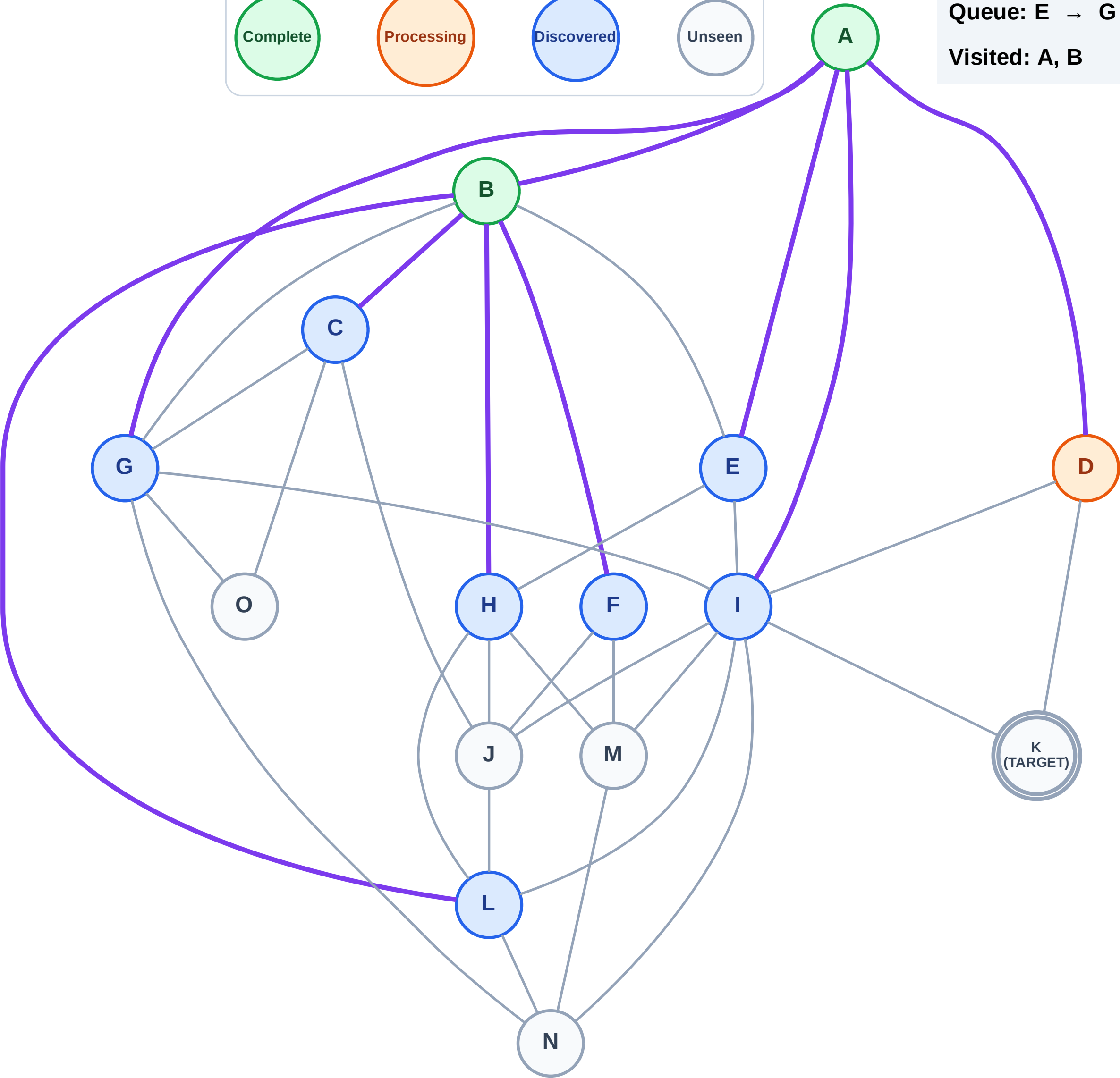
Processing

Discovered

Unseen

Queue: E → G → I → C → F → H → L

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

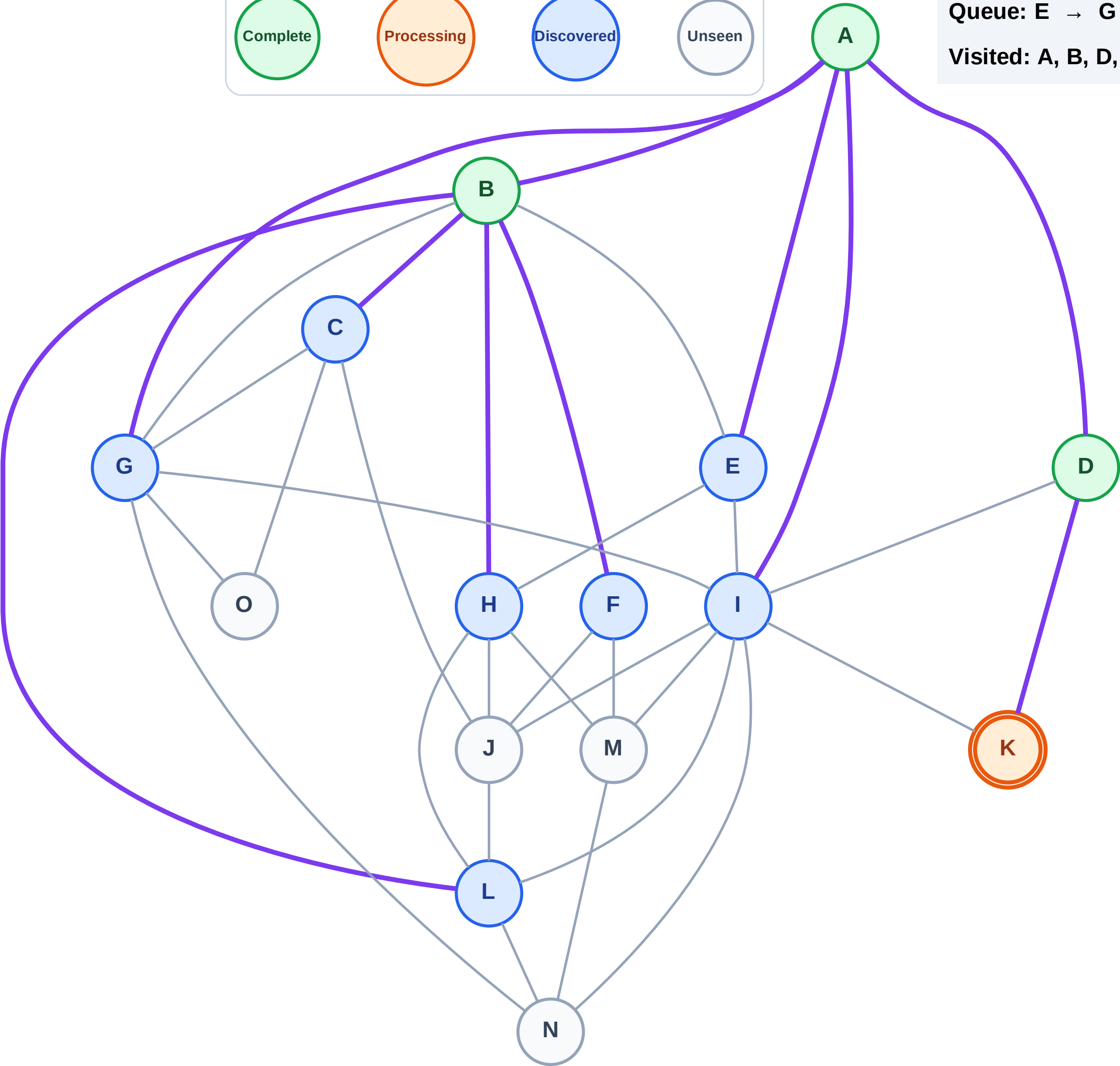
Processing

Discovered

Unseen

Queue: E → G → I → C → F → H → L

Visited: A, B, D, K



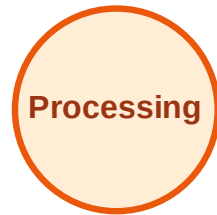
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



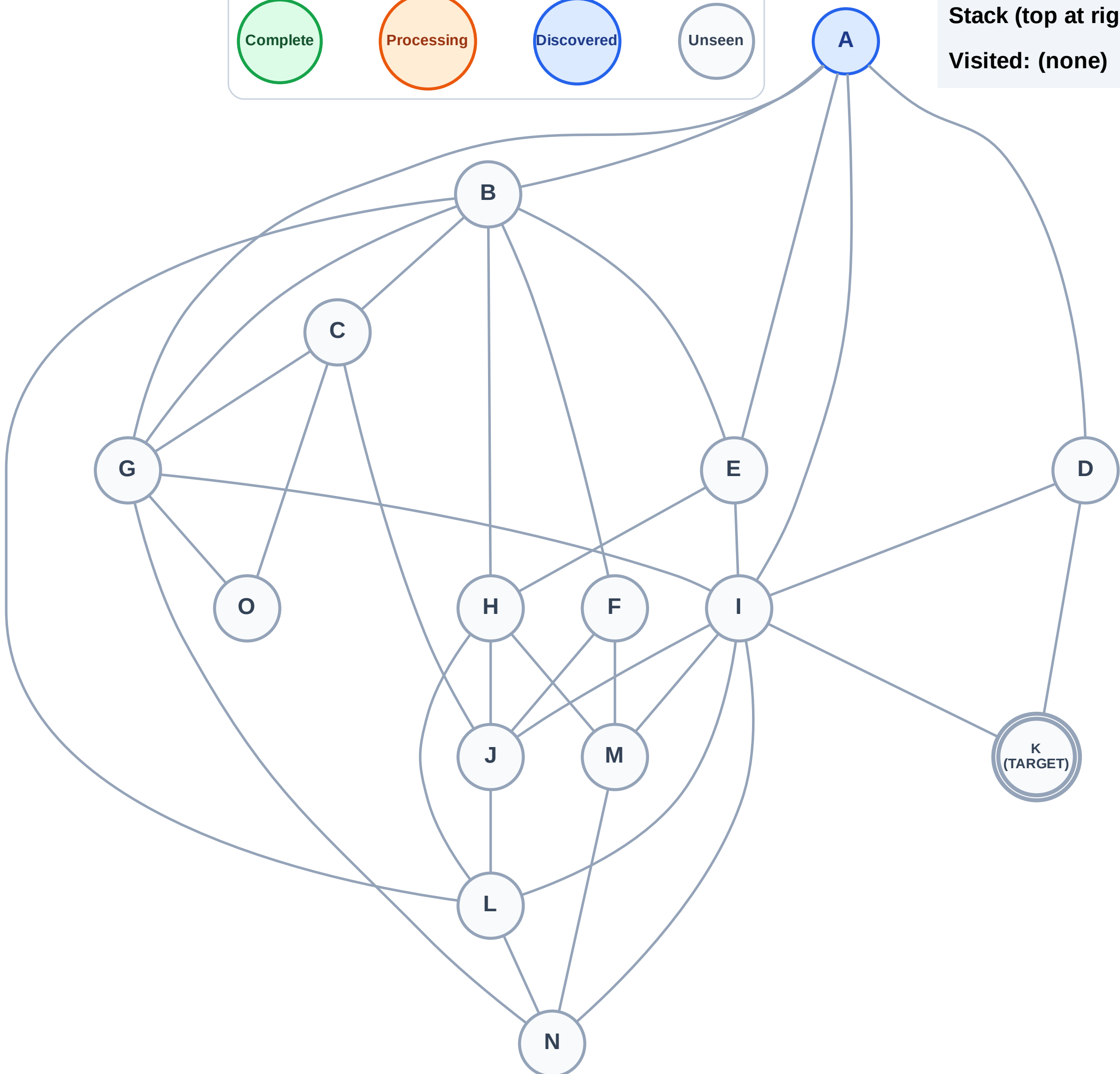
Discovered



Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

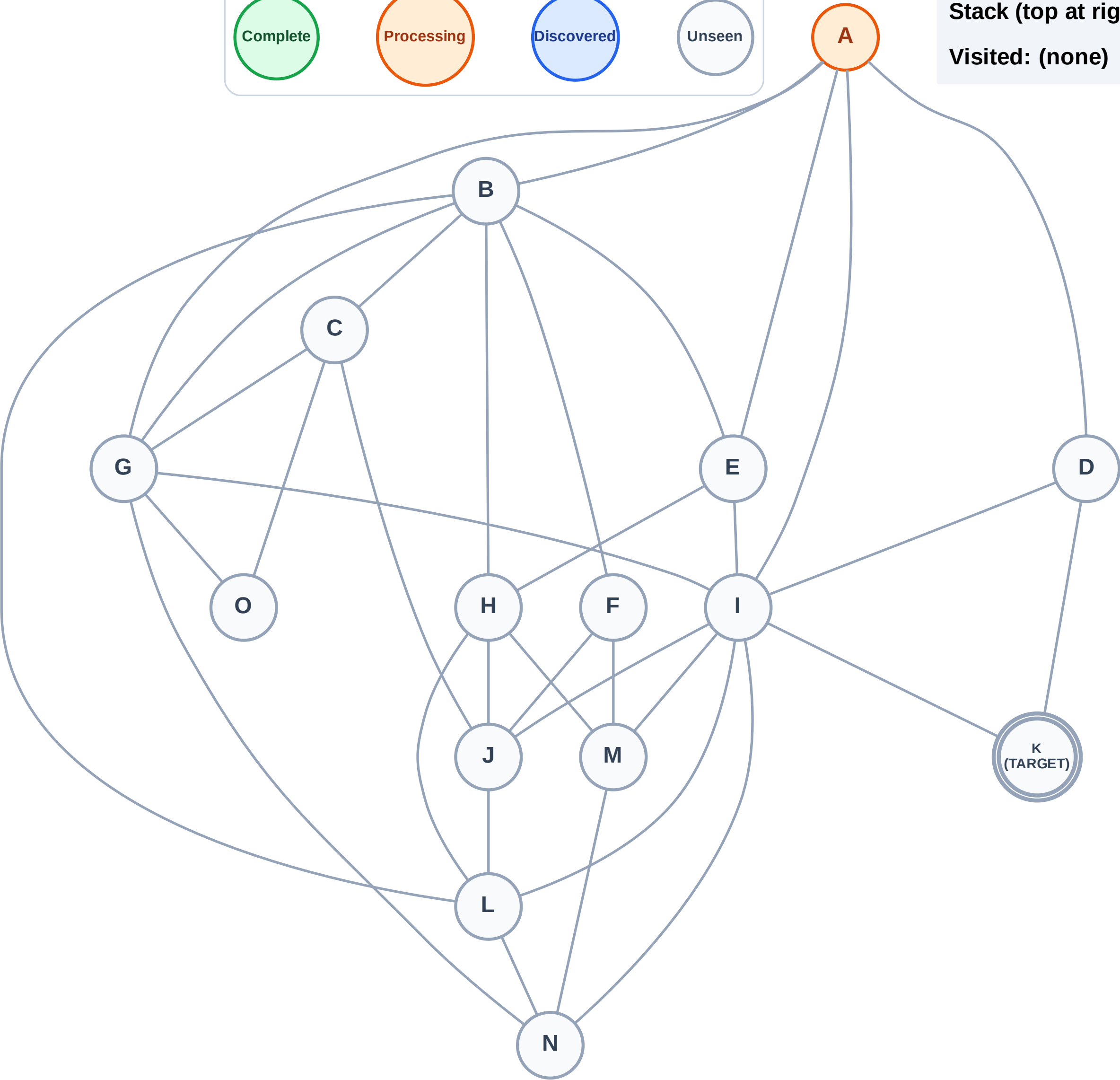
Complete

Processing

Discovered

Unseen

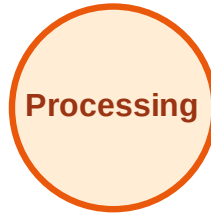
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

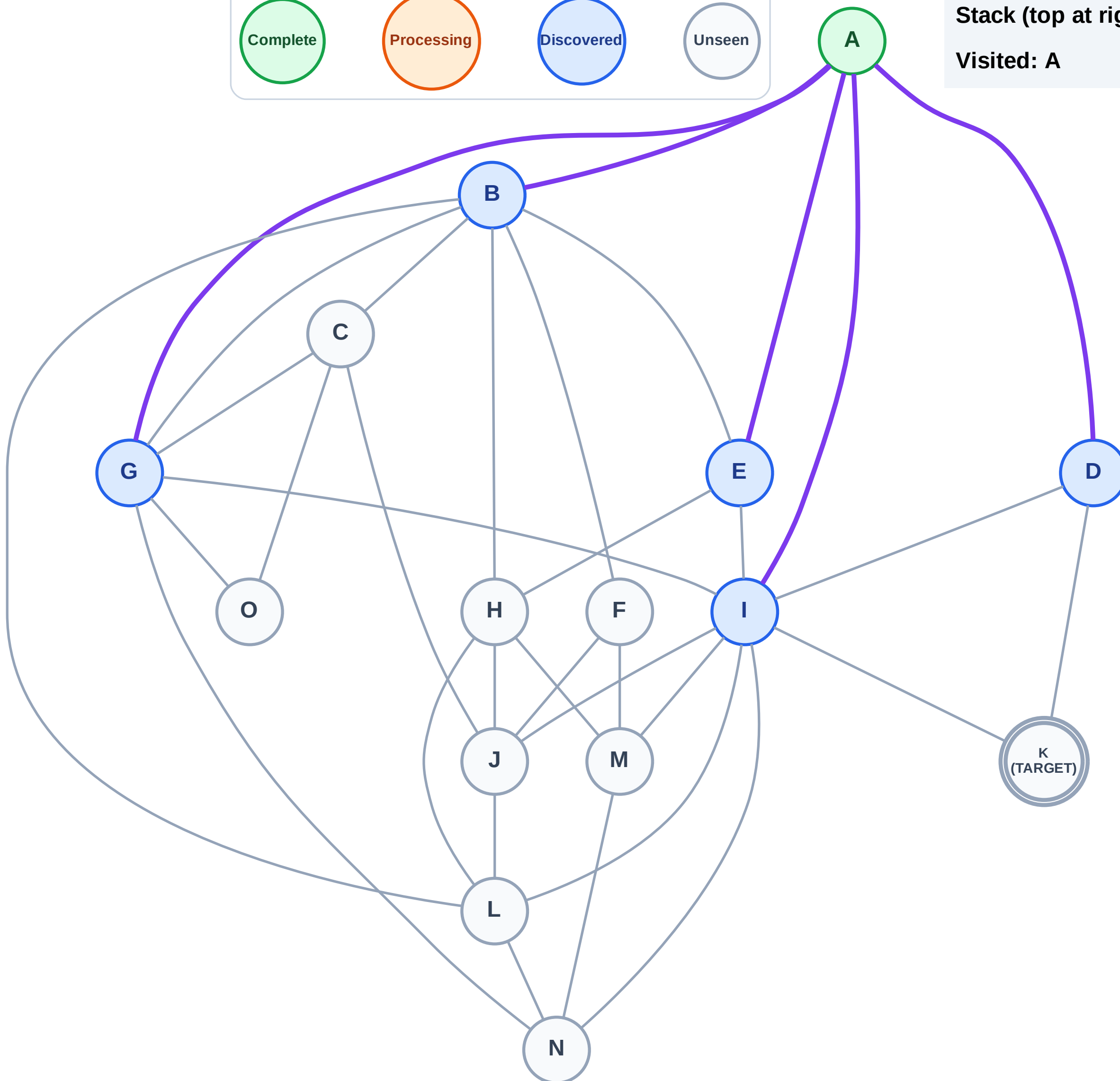
Step 3: Discover I, G, E, D, B from A

Legend



Stack (top at right): I | G | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

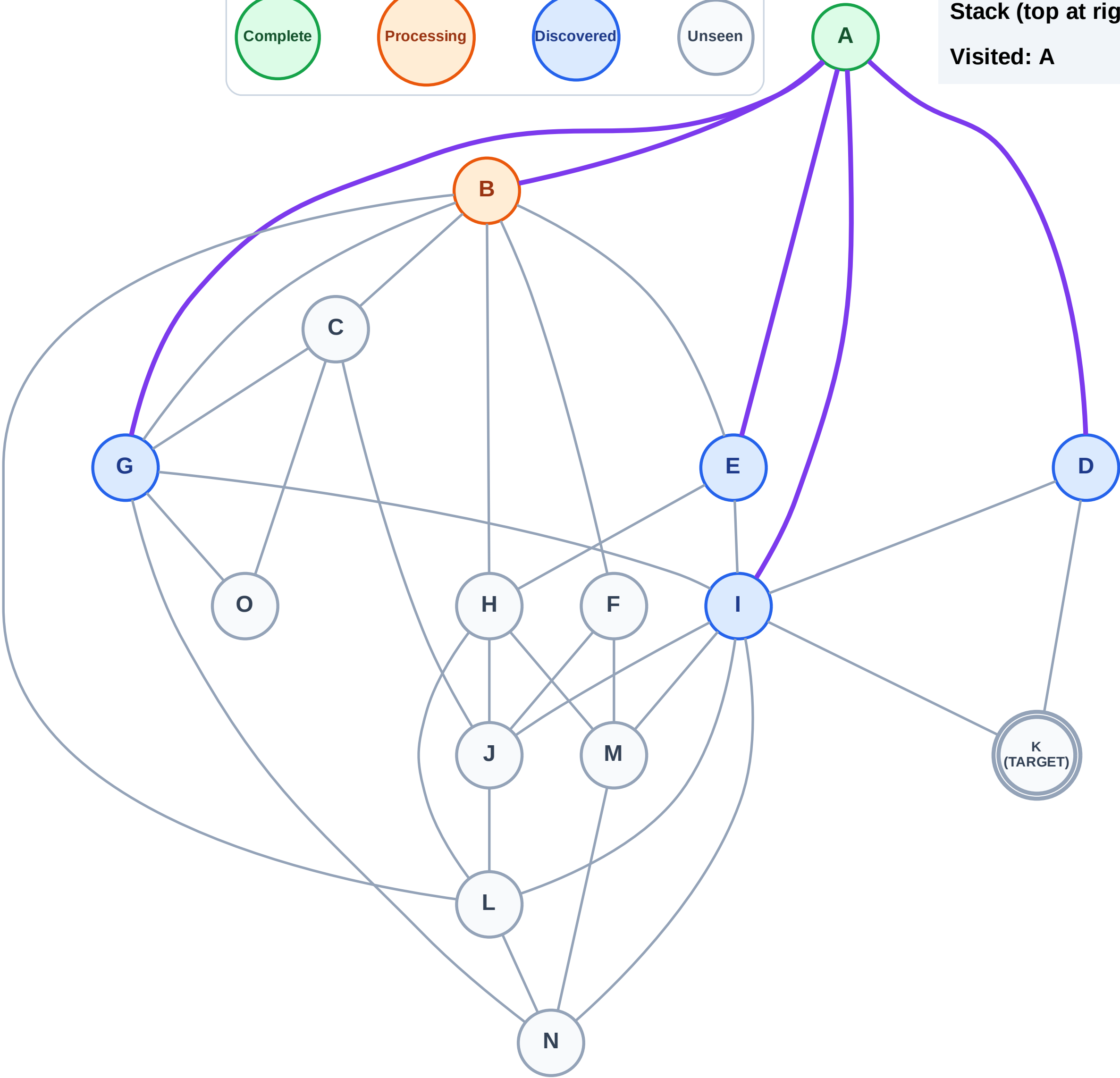
Processing

Discovered

Unseen

Stack (top at right): I | G | E | D

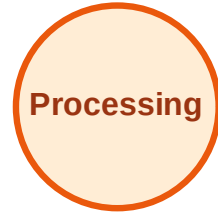
Visited: A



DFS Traversal

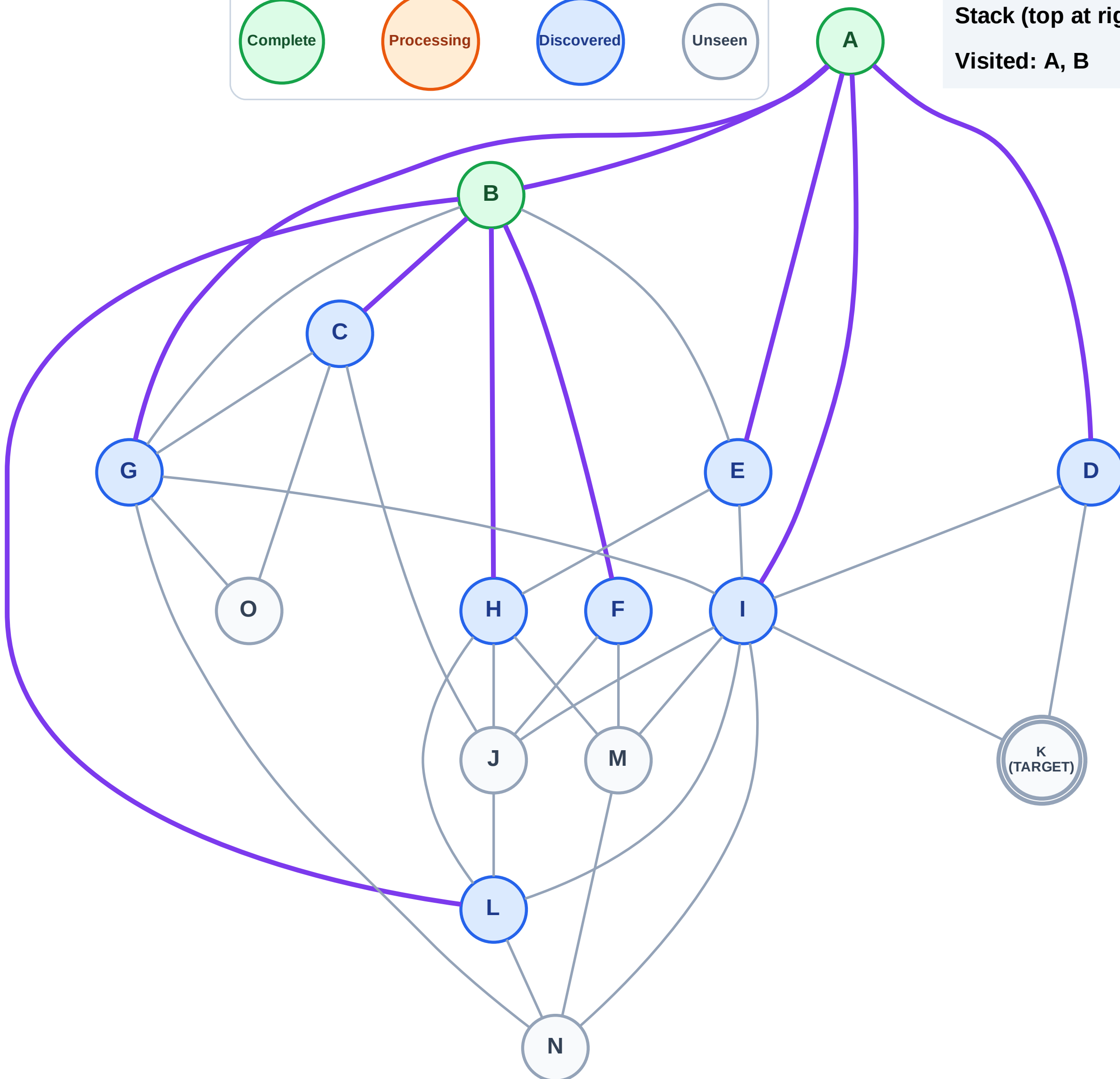
Step 5: Discover L, H, F, C from B

Legend



Stack (top at right): I | G | E | D | L | H | F | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

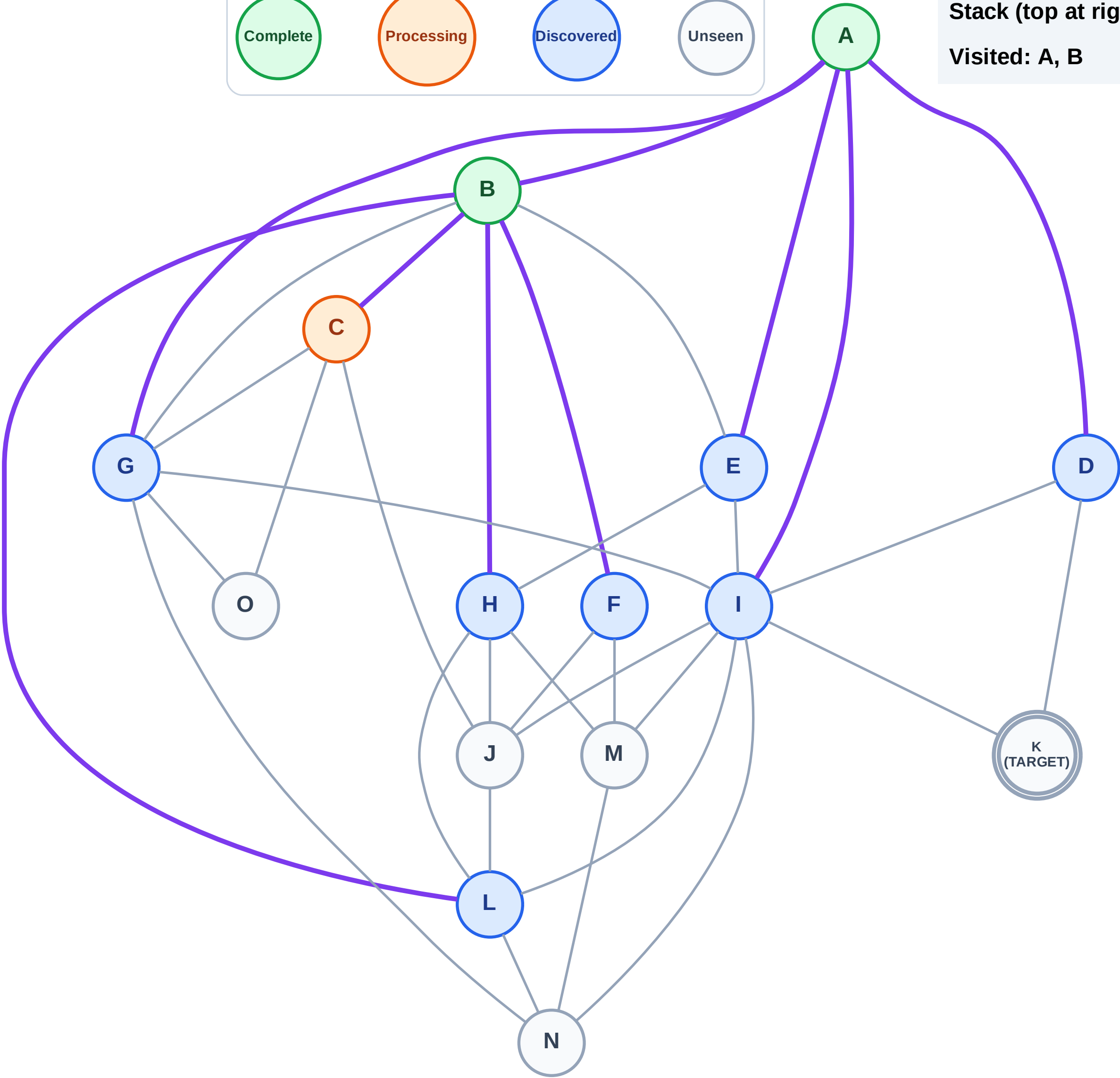
Processing

Discovered

Unseen

Stack (top at right): I | G | E | D | L | H | F

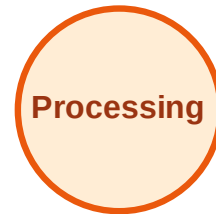
Visited: A, B



DFS Traversal

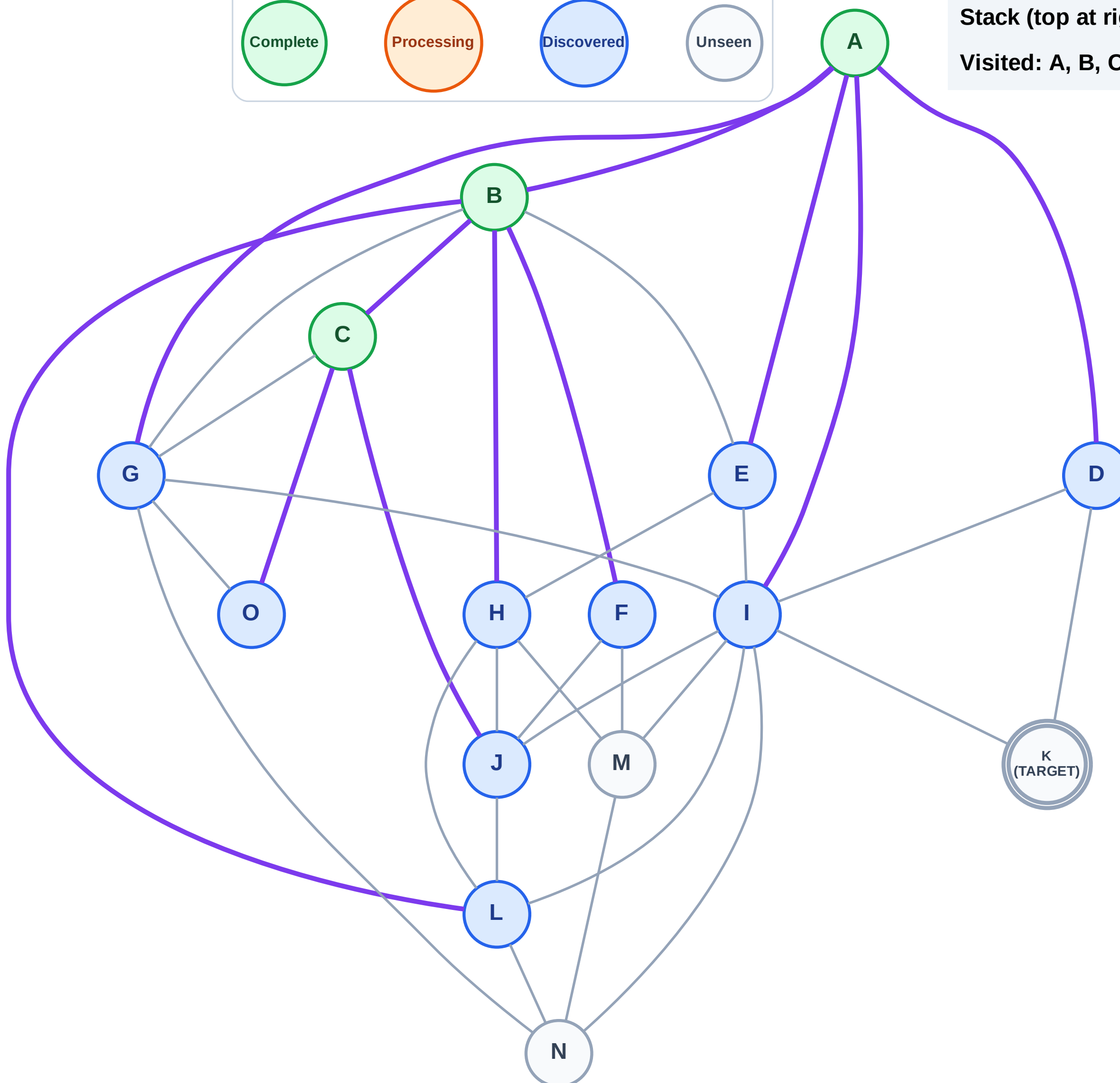
Step 7: Discover O, J from C

Legend



Stack (top at right): I | G | E | D | L | H | F | O | J

Visited: A, B, C



DFS Traversal

Step 8: Process node J

Legend

Complete

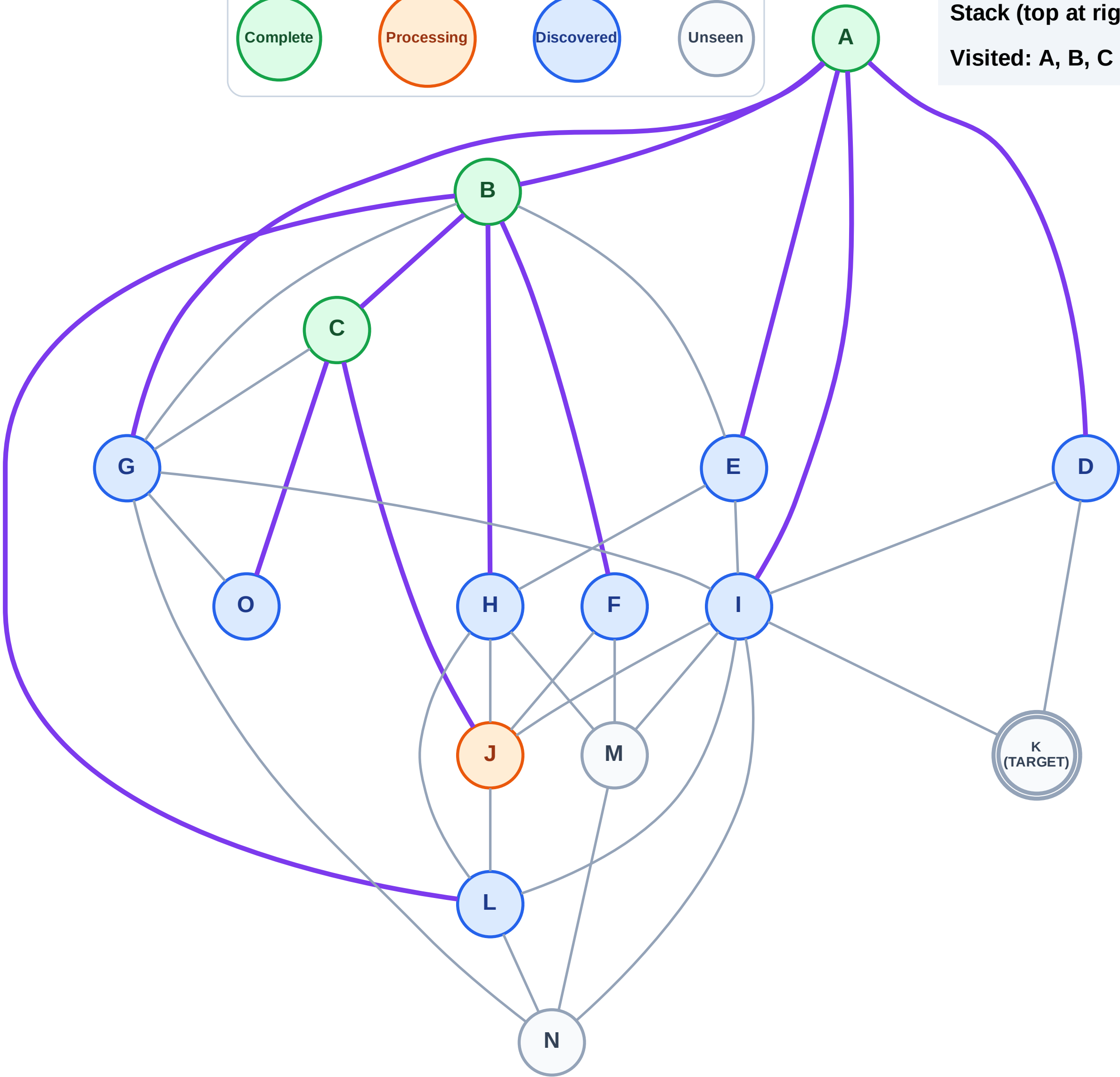
Processing

Discovered

Unseen

Stack (top at right): I | G | E | D | L | H | F | O

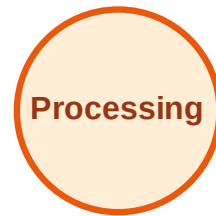
Visited: A, B, C



DFS Traversal

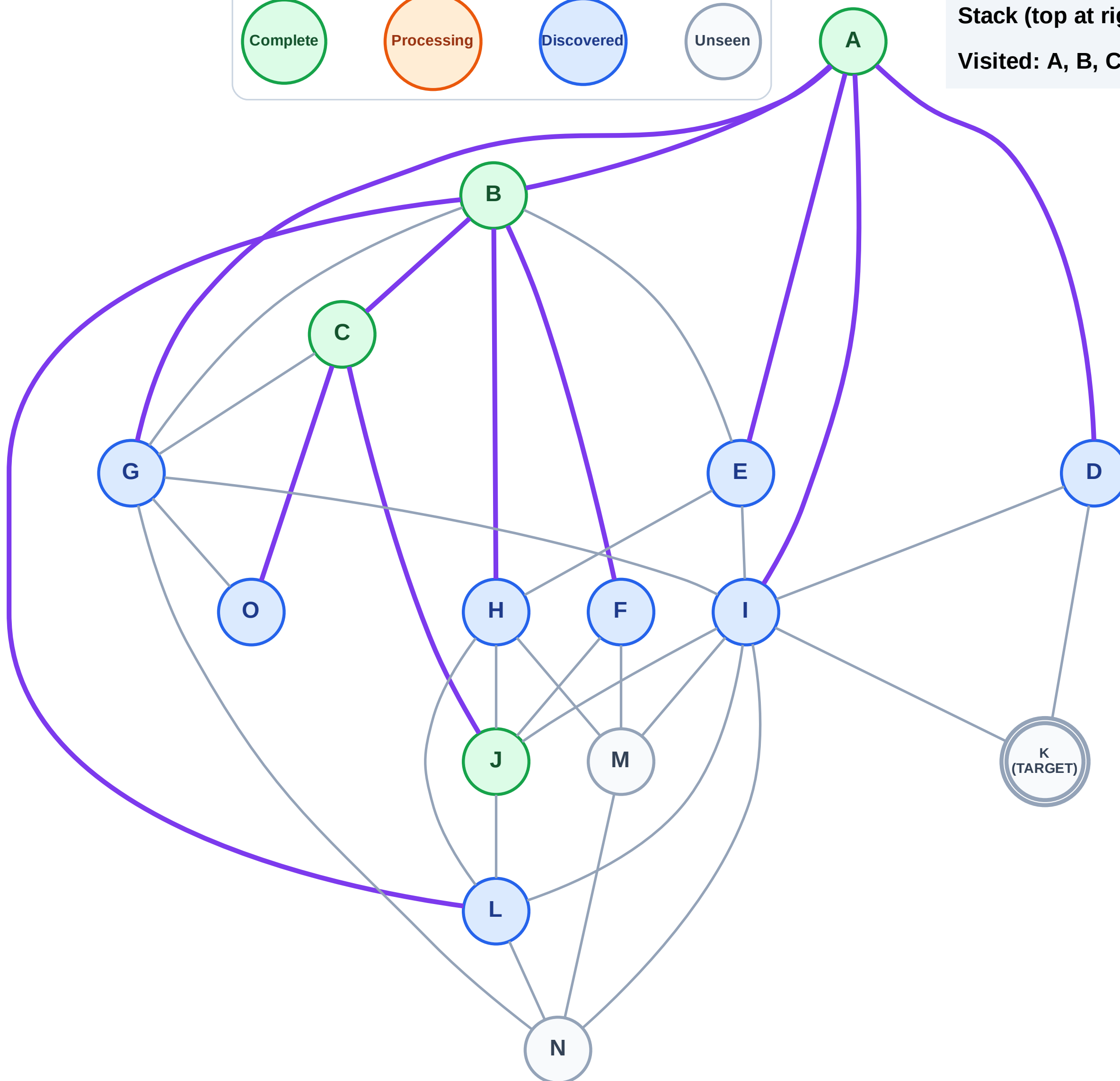
Step 9: No new neighbors from J

Legend



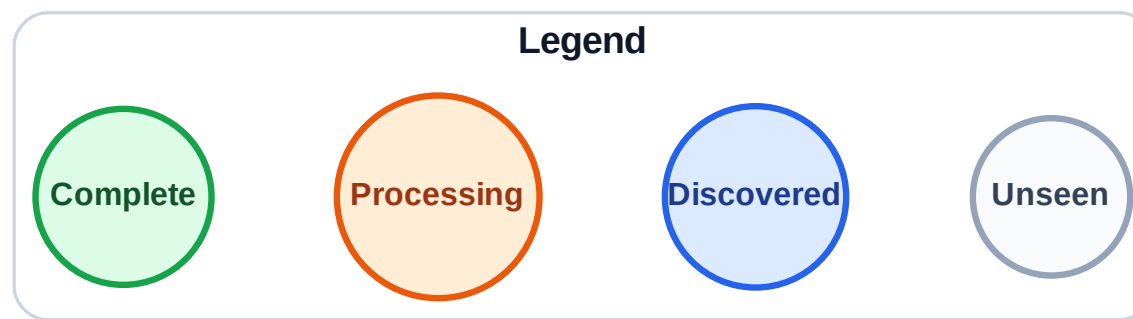
Stack (top at right): I | G | E | D | L | H | F | O

Visited: A, B, C, J



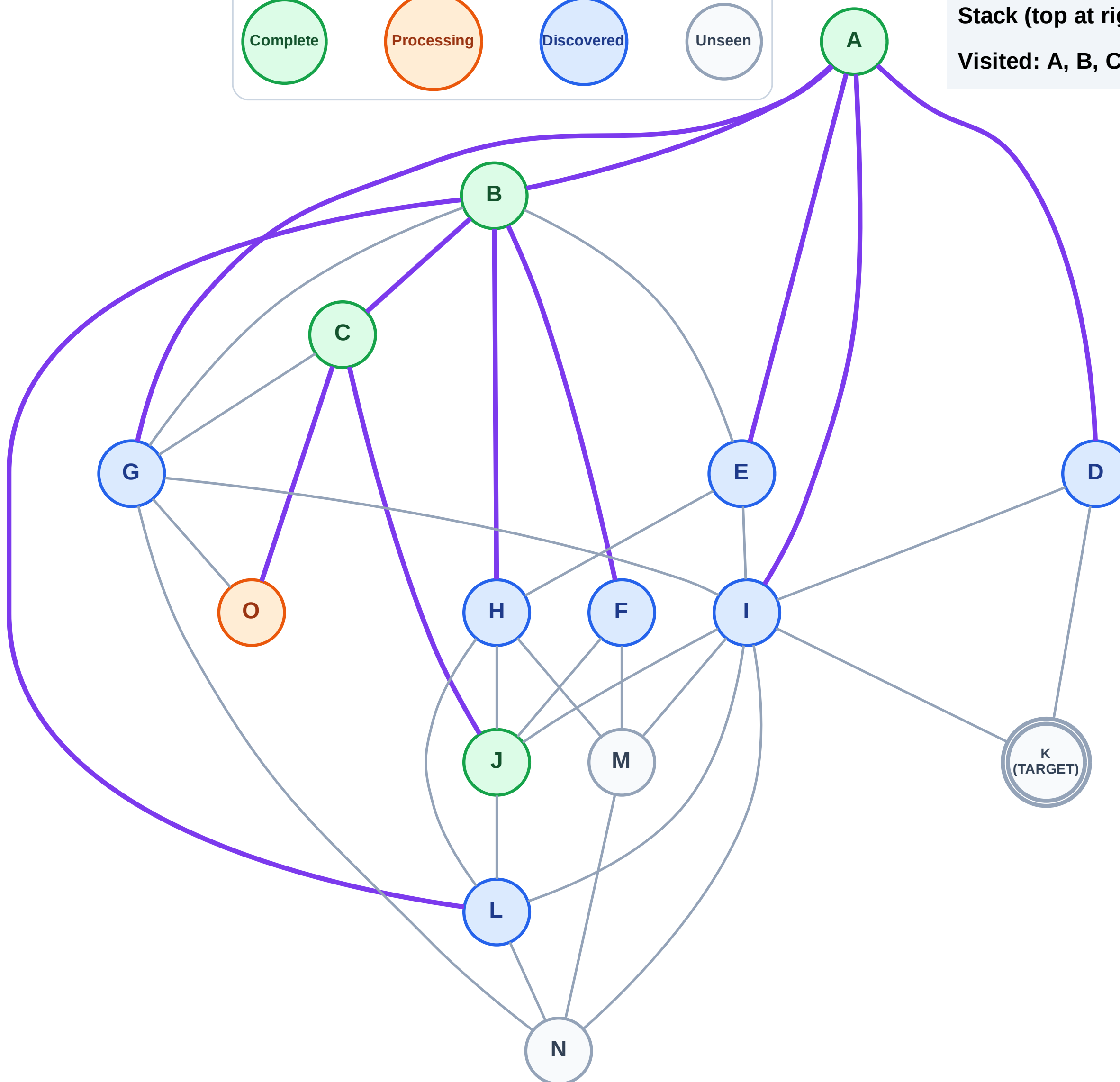
DFS Traversal

Step 10: Process node O



Stack (top at right): I | G | E | D | L | H | F

Visited: A, B, C, J



DFS Traversal

Step 11: No new neighbors from O

Legend

Complete

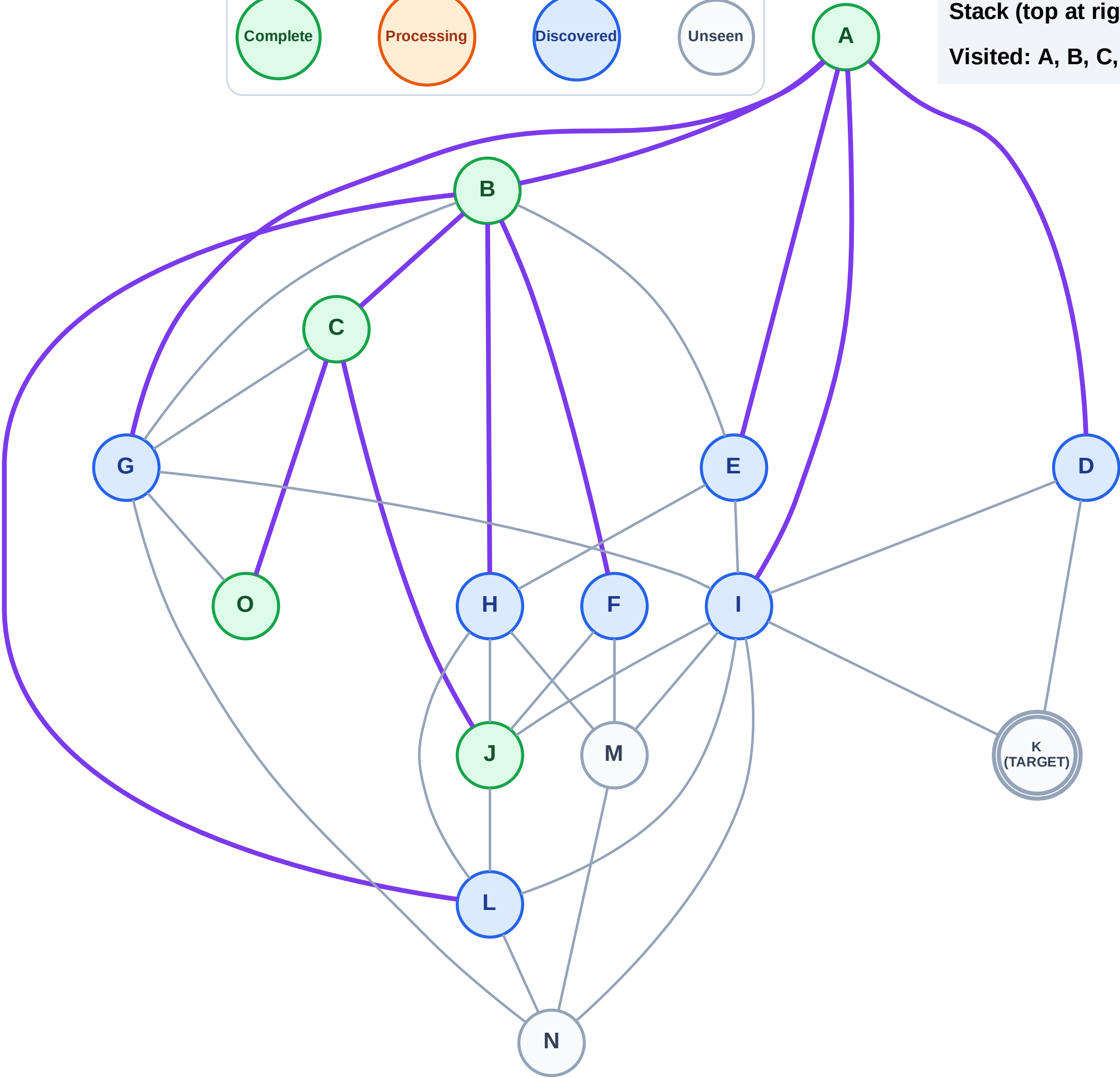
Processing

Discovered

Unseen

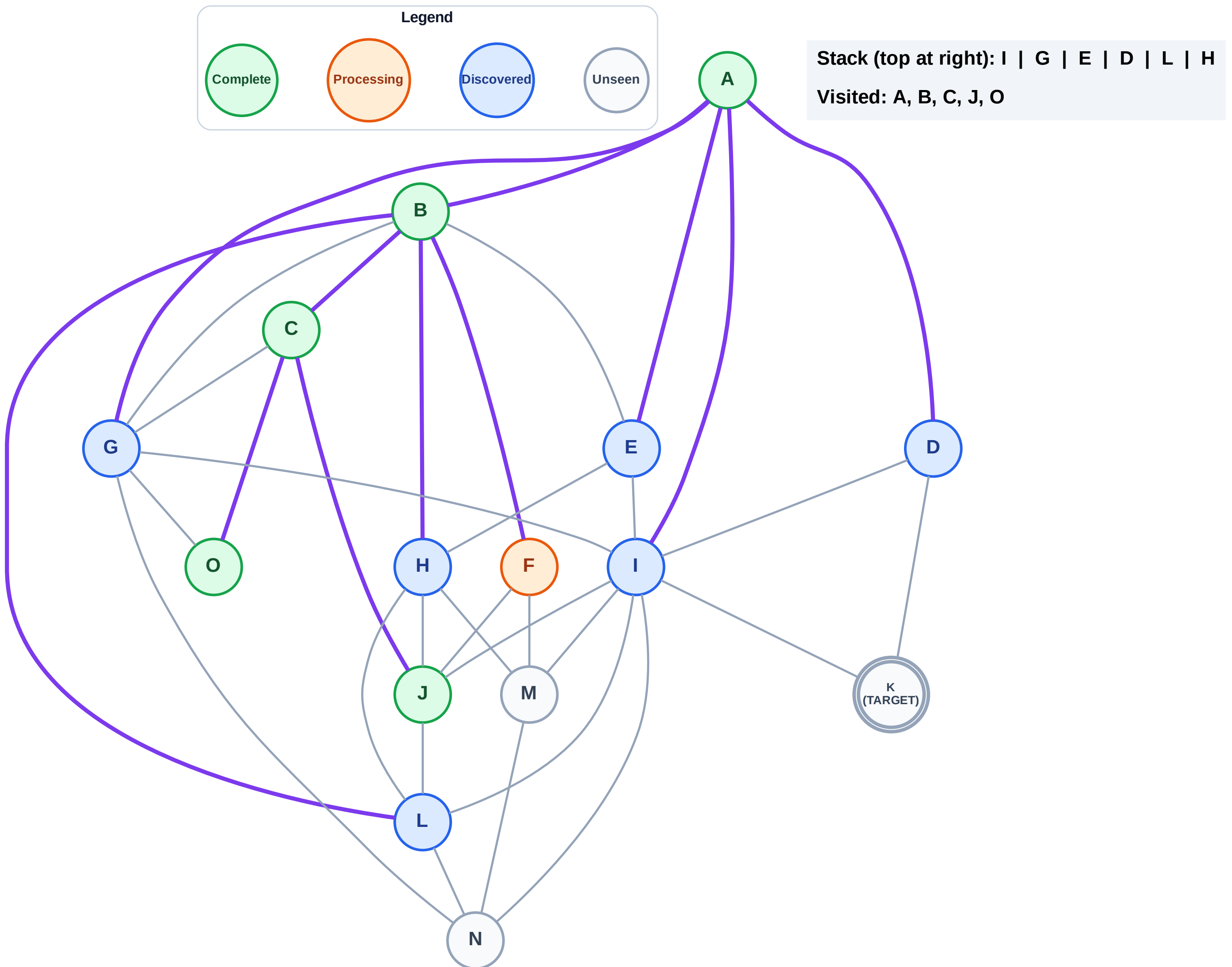
Stack (top at right): I | G | E | D | L | H | F

Visited: A, B, C, J, O



Step 12: Process node F

Step 12: Process node F



DFS Traversal

Step 13: Discover M from F

Legend



Complete



Processing



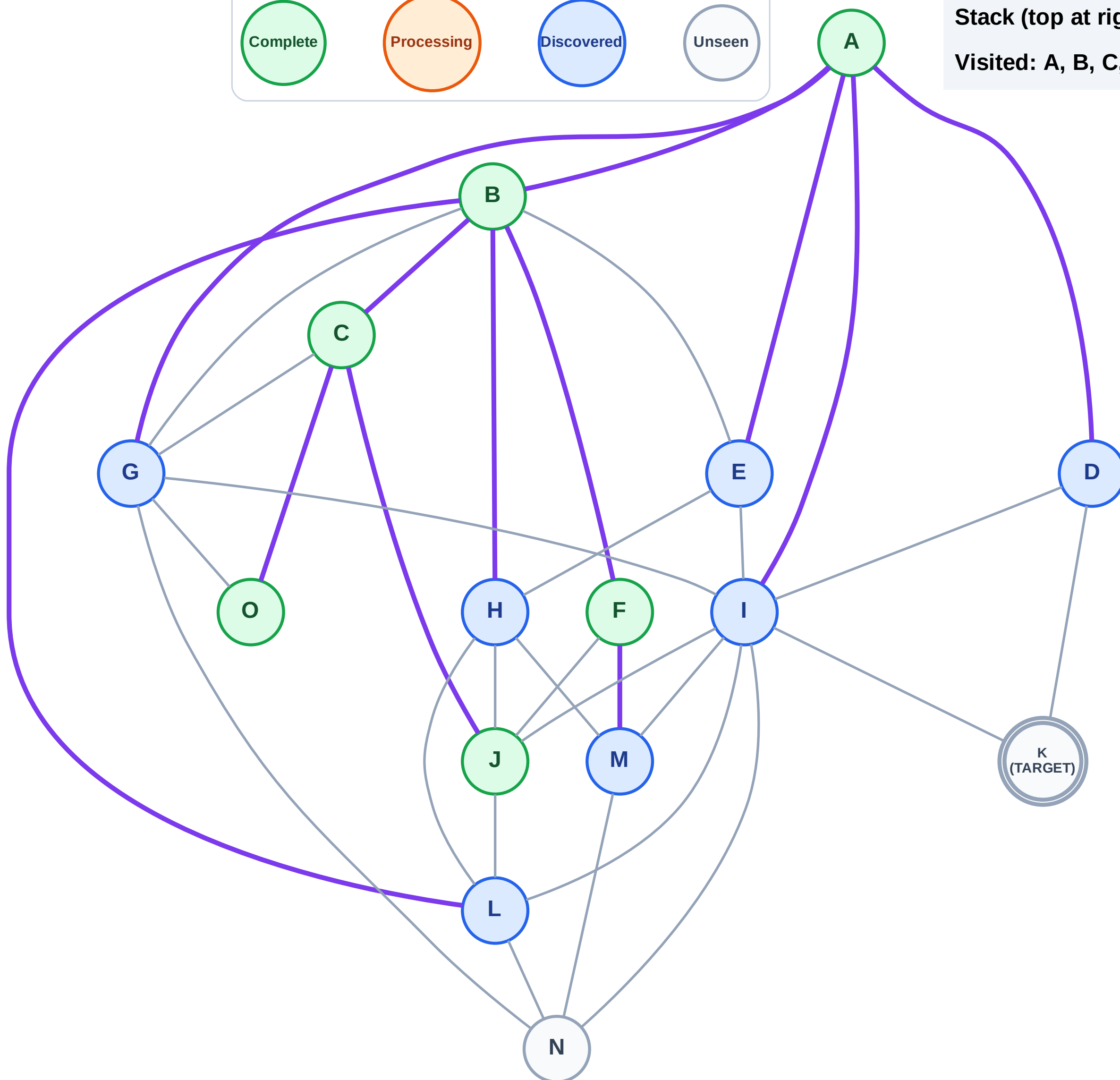
Discovered



Unseen

Stack (top at right): I | G | E | D | L | H | M

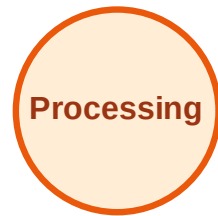
Visited: A, B, C, F, J, O



DFS Traversal

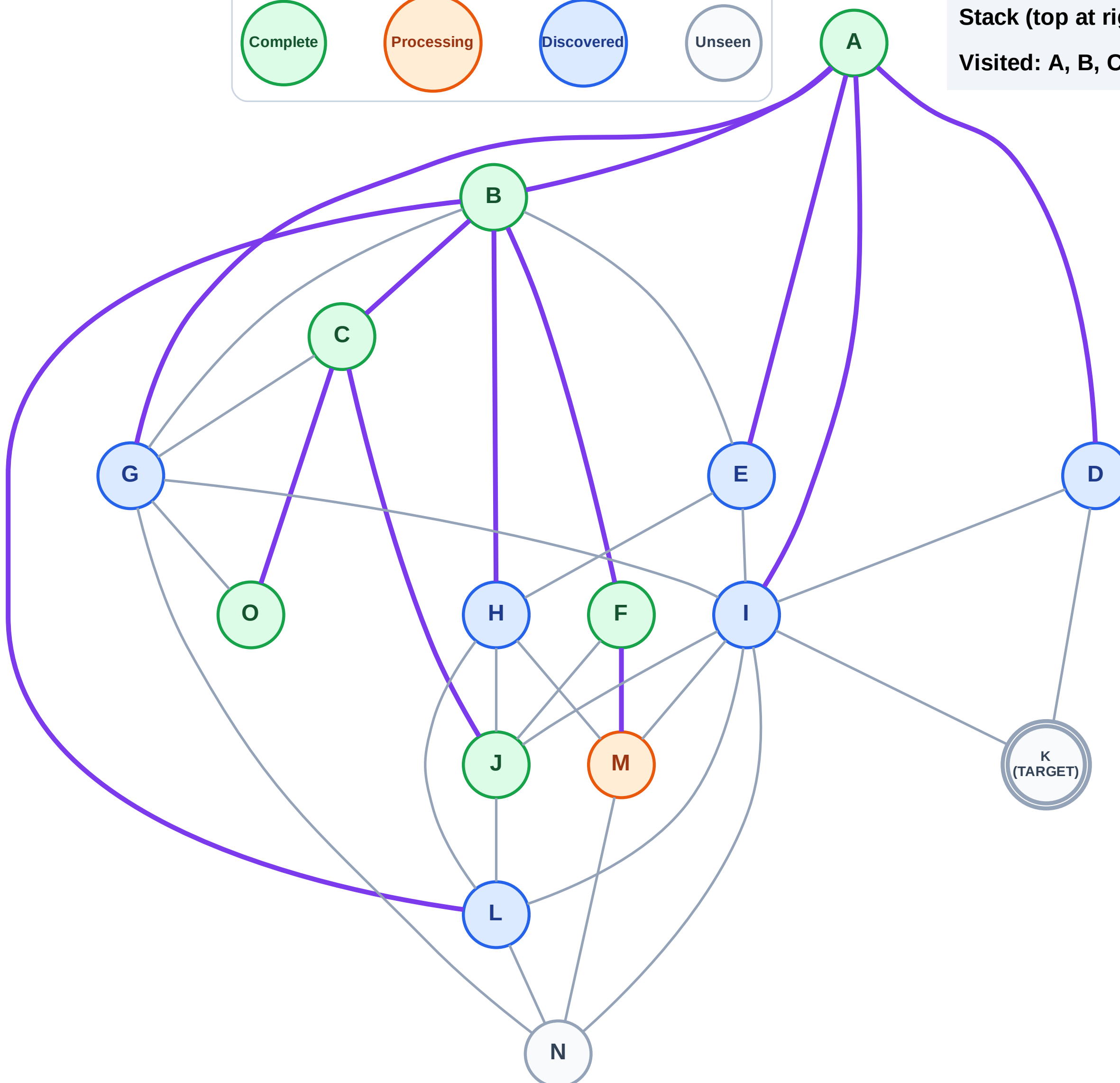
Step 14: Process node M

Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, F, J, O



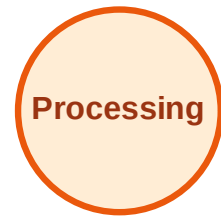
DFS Traversal

Step 15: Discover N from M

Legend



Complete



Processing



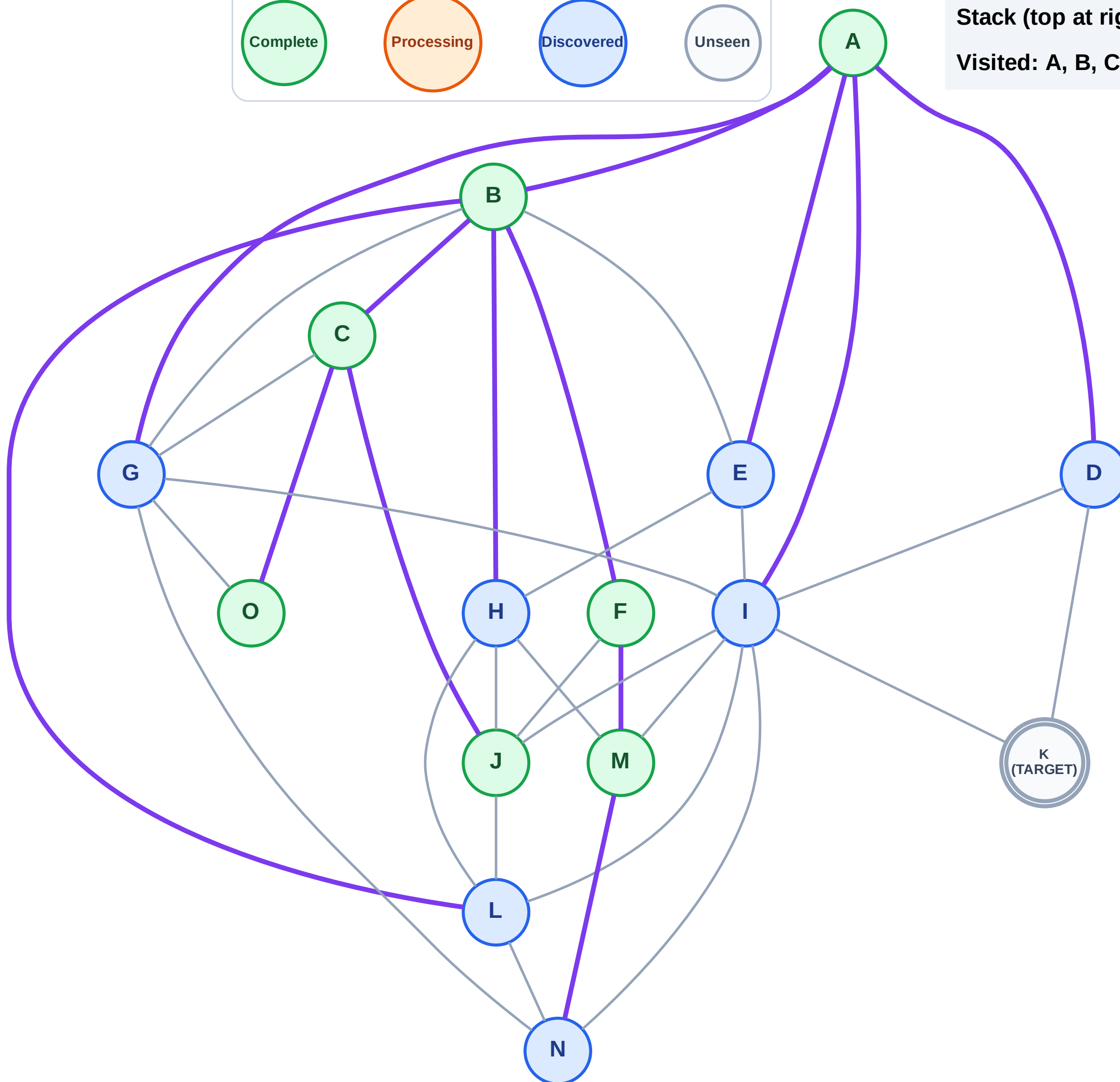
Discovered



Unseen

Stack (top at right): I | G | E | D | L | H | N

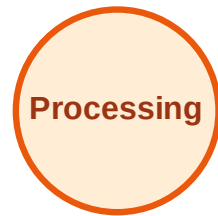
Visited: A, B, C, F, J, M, O



DFS Traversal

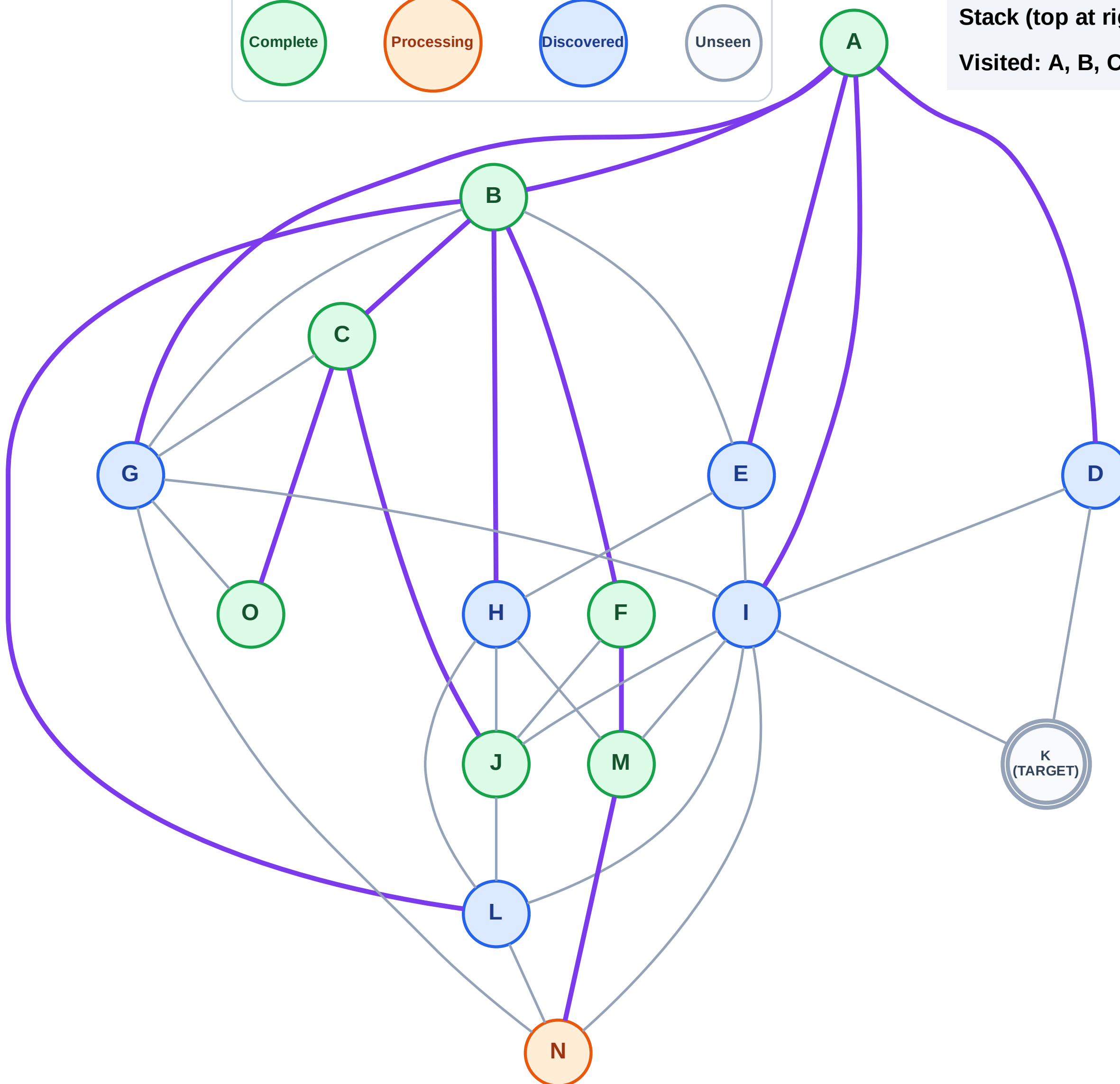
Step 16: Process node N

Legend



Stack (top at right): I | G | E | D | L | H

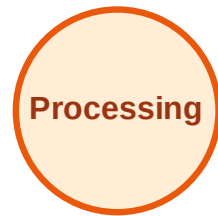
Visited: A, B, C, F, J, M, O



DFS Traversal

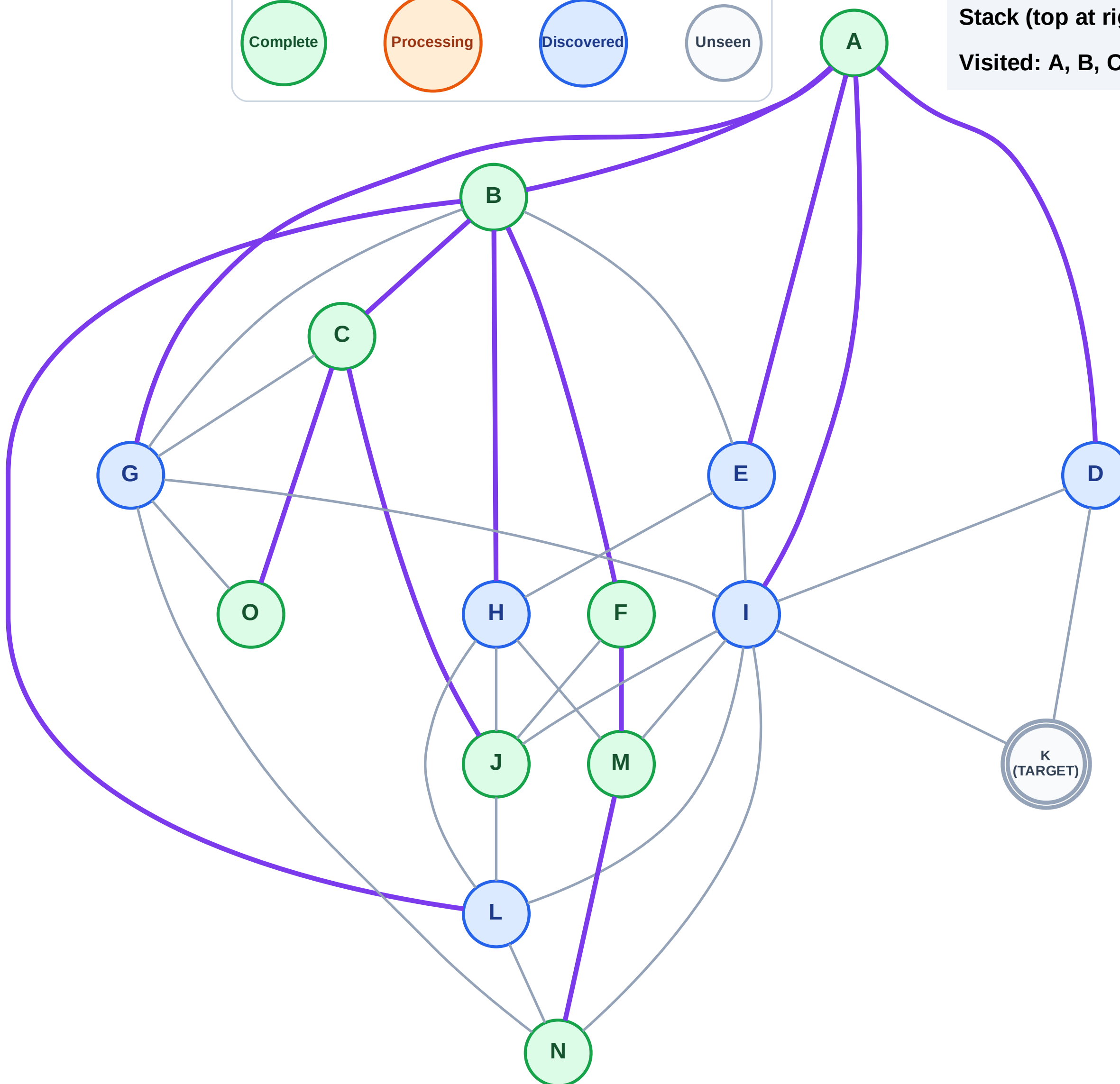
Step 17: No new neighbors from N

Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, F, J, M, N, O



DFS Traversal

Step 18: Process node H

Legend



Complete



Processing



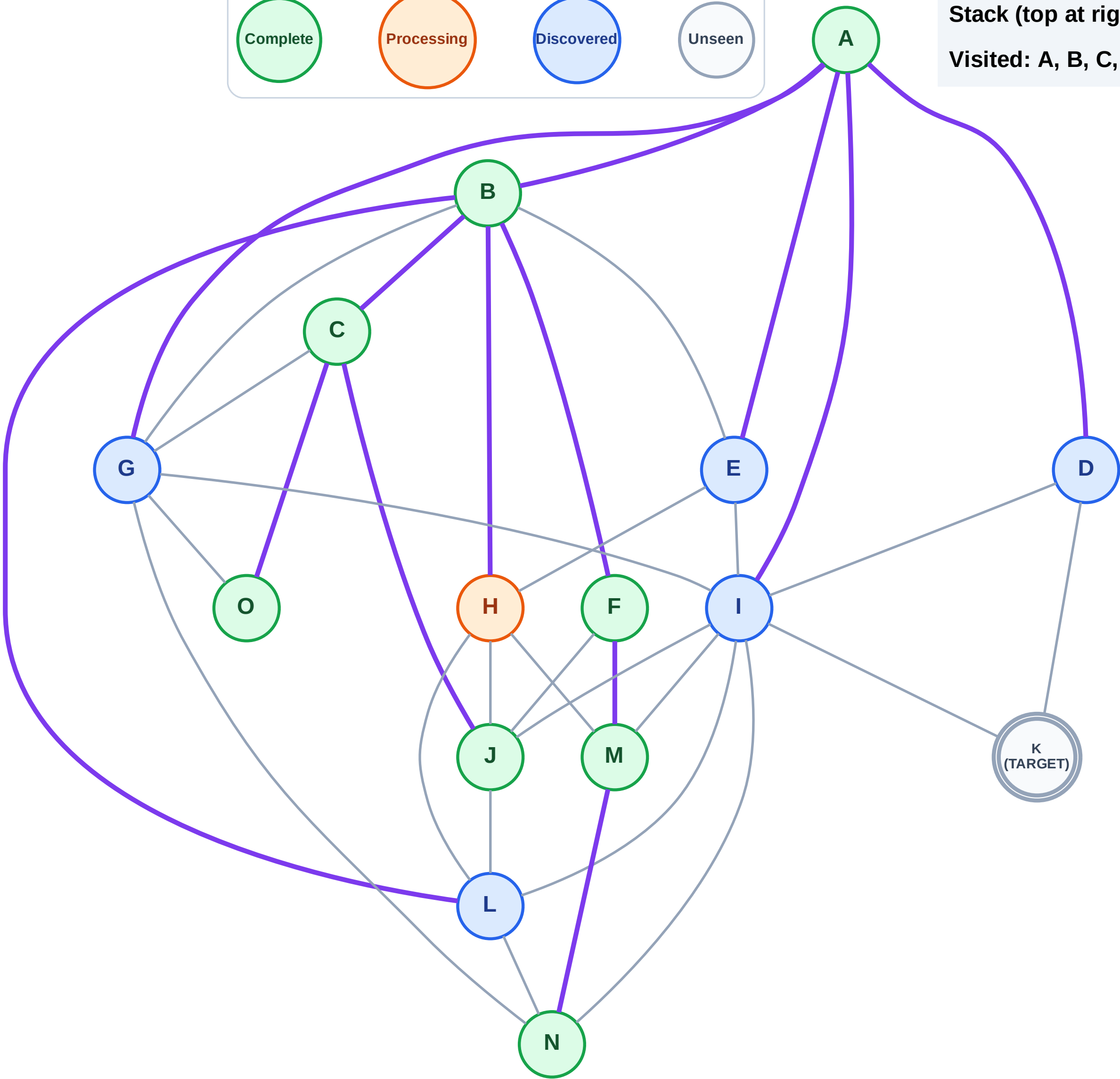
Discovered



Unseen

Stack (top at right): I | G | E | D | L

Visited: A, B, C, F, J, M, N, O



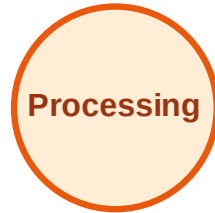
DFS Traversal

Step 19: No new neighbors from H

Legend



Complete



Processing



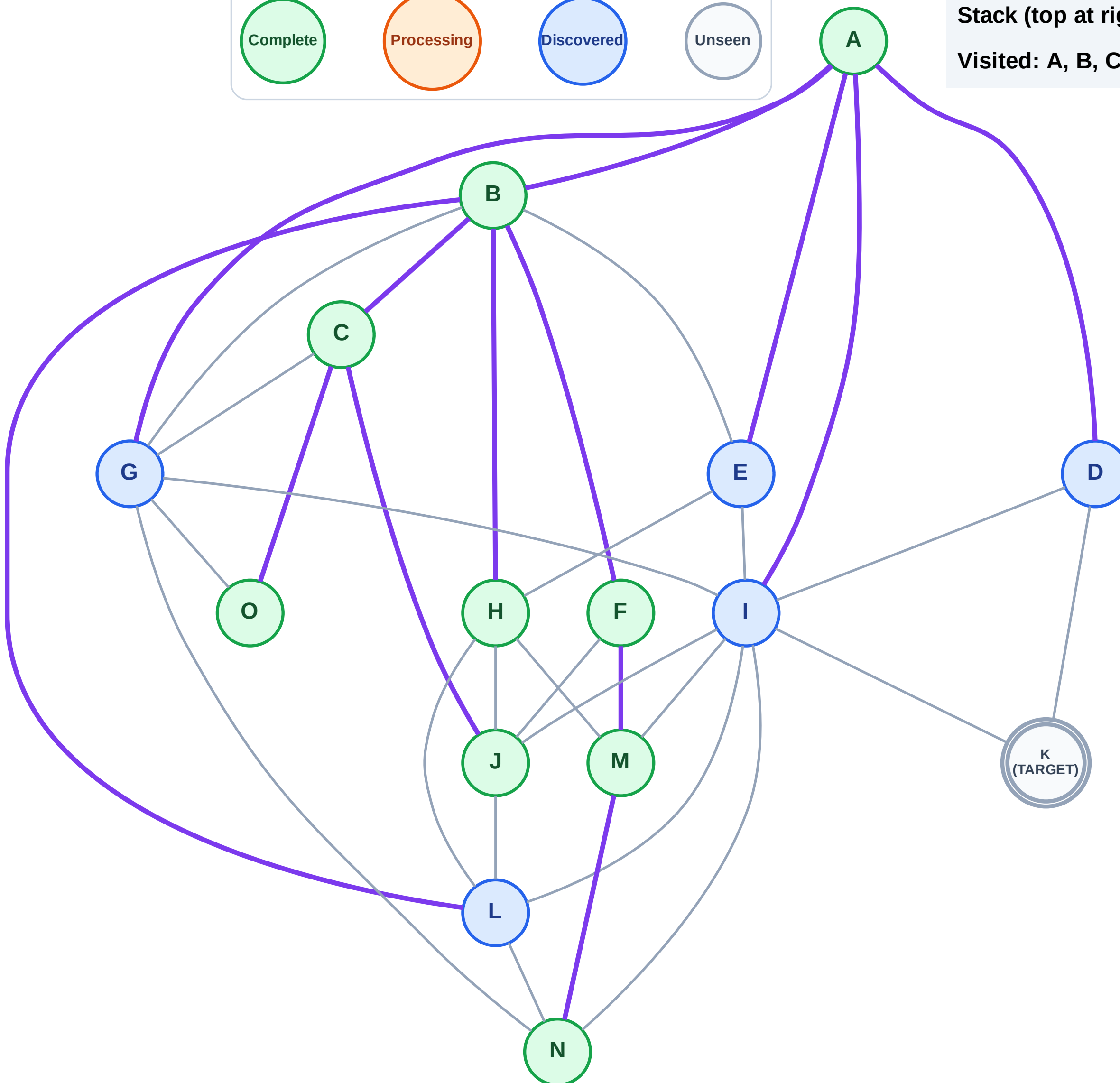
Discovered



Unseen

Stack (top at right): I | G | E | D | L

Visited: A, B, C, F, H, J, M, N, O



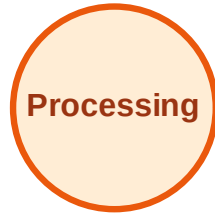
DFS Traversal

Step 20: Process node L

Legend



Complete



Processing



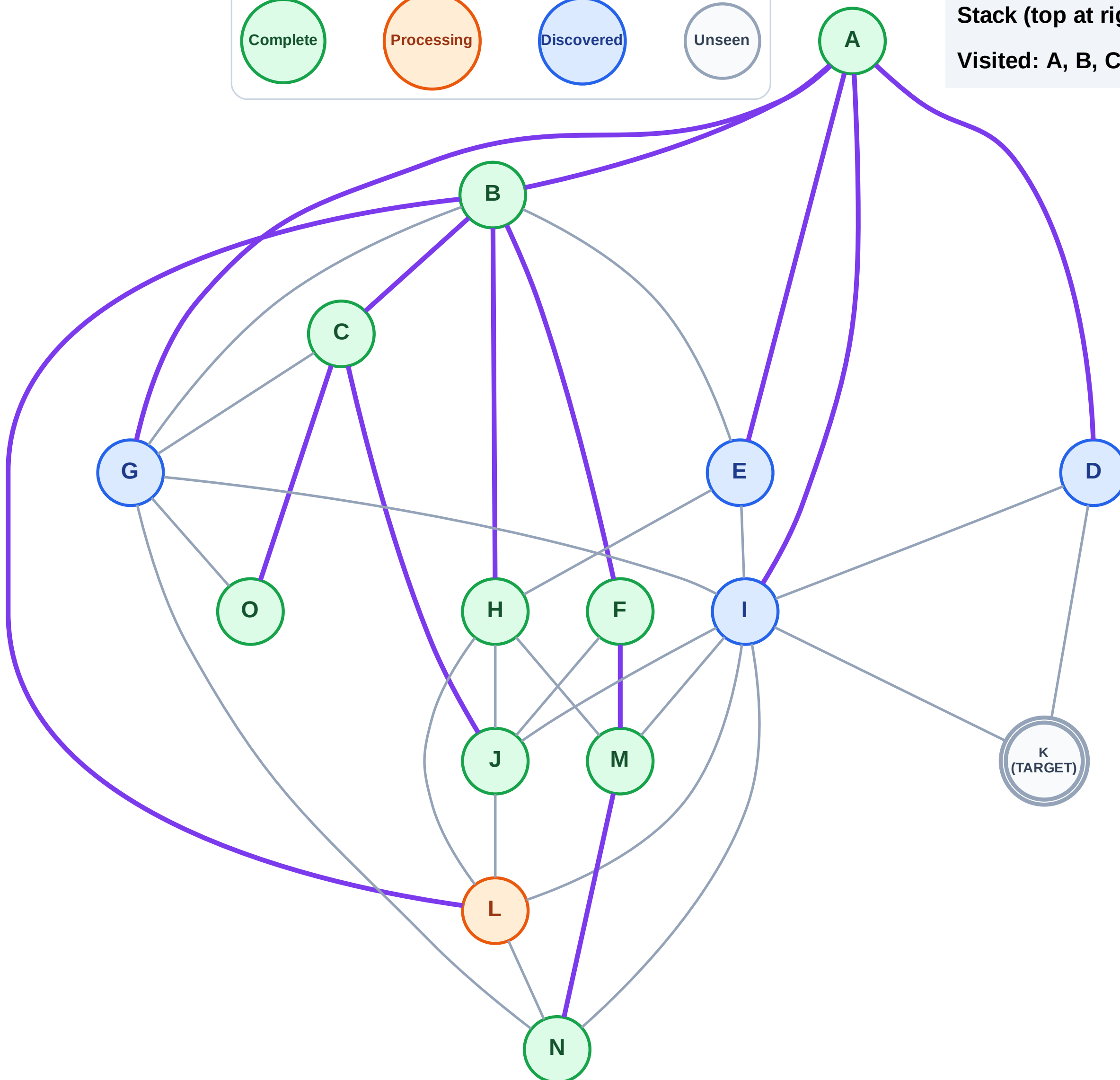
Discovered



Unseen

Stack (top at right): I | G | E | D

Visited: A, B, C, F, H, J, M, N, O



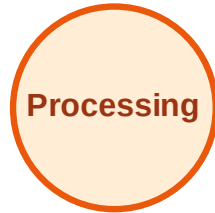
DFS Traversal

Step 21: No new neighbors from L

Legend



Complete



Processing



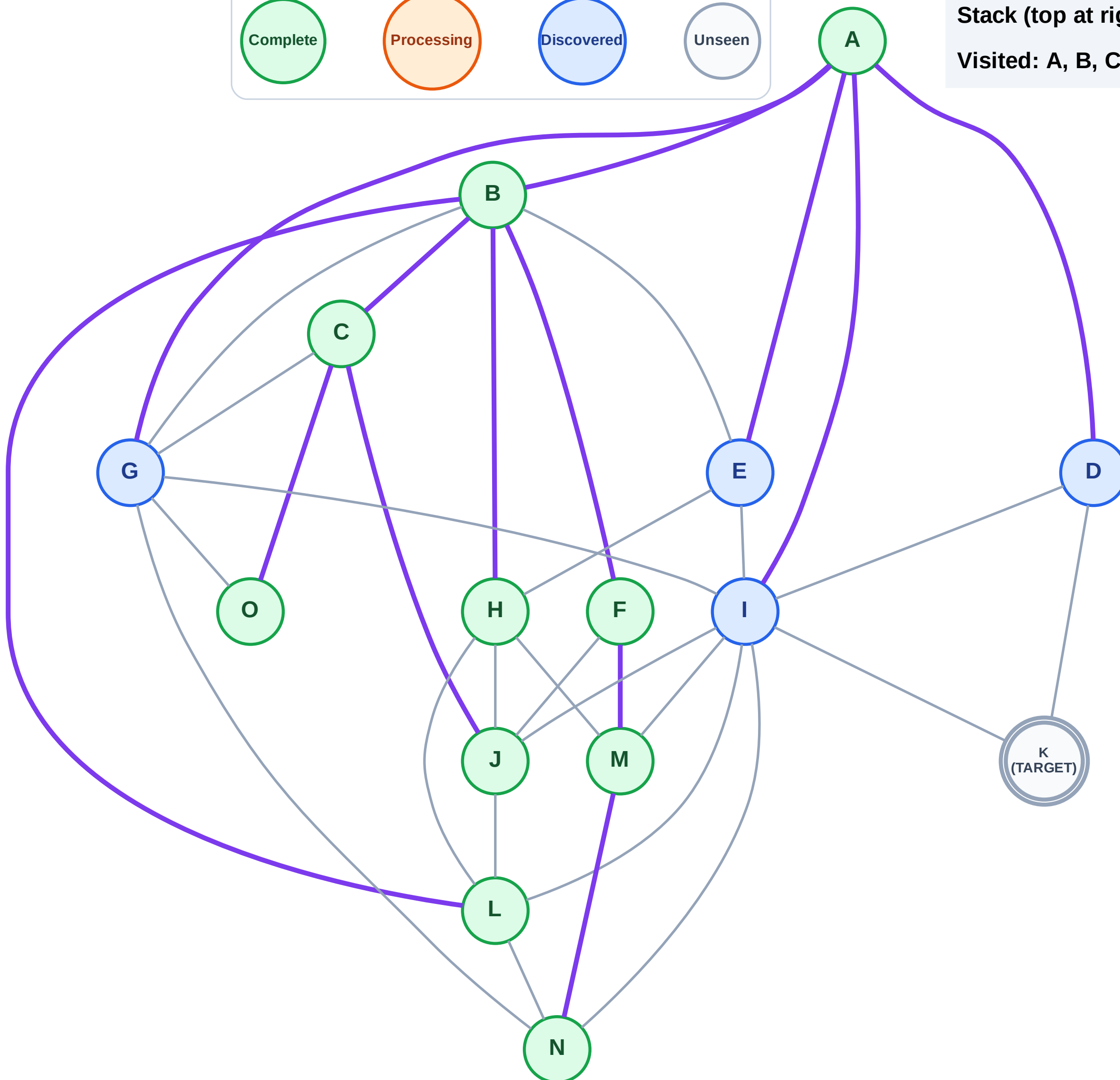
Discovered



Unseen

Stack (top at right): I | G | E | D

Visited: A, B, C, F, H, J, L, M, N, O



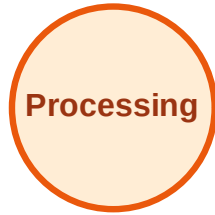
DFS Traversal

Step 22: Process node D

Legend



Complete



Processing

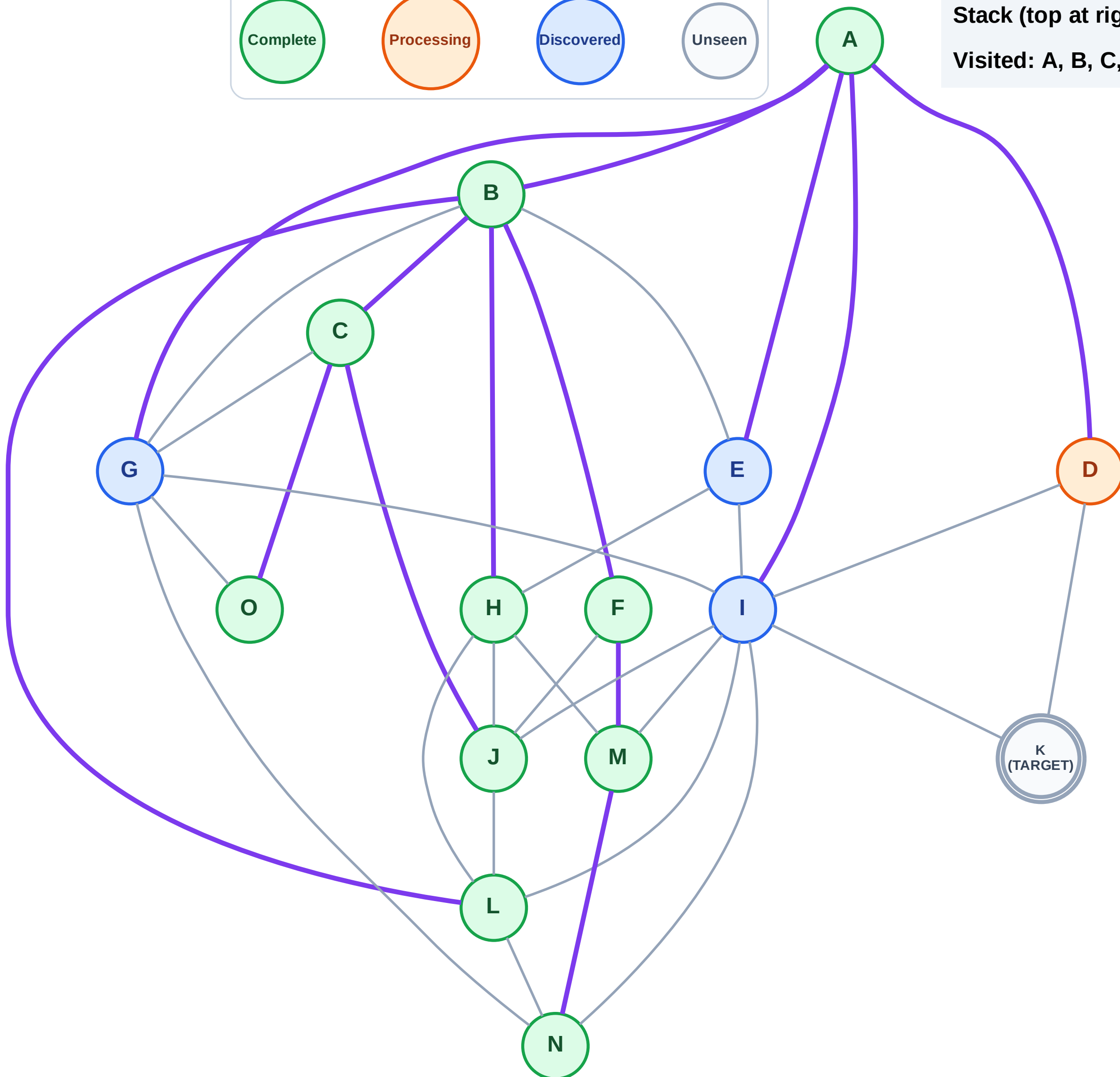


Discovered



Unseen

Stack (top at right): I | G | E
Visited: A, B, C, F, H, J, L, M, N, O



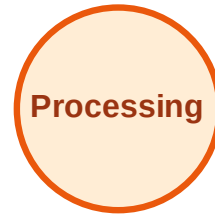
DFS Traversal

Step 23: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): I | G | E

Visited: A, B, C, D, F, H, J, K, L, M, N, O

